

✉ For correspondence:

funabih@rockefeller.edu

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Impacts of DNA Methylation on H2A.Z Deposition and Nucleosome Stability

Rochelle M Shih¹, Yasuhiro Arimura^{1,2}, Hide A Konishi¹, Hironori Funabiki¹ ✉

¹Laboratory of Chromosome and Cell Biology, The Rockefeller University, New York, United States • ²Basic Sciences Division, Fred Hutchinson Cancer Center, Seattle, United States

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This study provides **valuable** mechanistic insight into the mutually exclusive distributions of the histone variant H2A.Z and DNA methylation by testing two hypotheses: (i) that DNA methylation destabilizes H2A.Z nucleosomes, thereby preventing H2A.Z retention, and (ii) that DNA methylation suppresses H2A.Z deposition by ATP-dependent chromatin remodelling complexes. Through a series of well-designed and carefully executed experiments, findings are presented in support of both hypotheses. However, the evidence in support of either hypothesis is **incomplete**, so that the proposed mechanisms underlying the enrichment of H2A.Z on unmethylated DNA remain somewhat speculative.

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Abstract

In eukaryotes with DNA methylation, the histone variant H2A.Z and DNA methylation are maintained in mutually exclusive sections of the genome. How this antagonism is established, however, remains an open question. Here, we examined the impacts of DNA methylation on both the intrinsic stability of H2A.Z nucleosomes and chaperone-mediated H2A.Z deposition. Cryo-EM and endonuclease accessibility analyses show that H2A.Z nucleosomes with methylated DNA are more open and accessible compared to their unmethylated counterparts. In *Xenopus laevis*, H2A.Z preferentially associates with unmethylated DNA in both the fibroblast cell line XTC-2 and sperm pronuclei formed in the transcriptionally silent egg extract. The proportion of H2A.Z that colocalizes with methylated DNA, however, is higher in sperm pronuclei than in XTC-2. By monitoring nucleosome assembly on synthetic DNA constructs in *Xenopus* egg extracts, we find that the H2A.Z bias for unmethylated substrates is dependent on the SRCAP complex, the major H2A.Z deposition chaperone. Consistently, recruitment of the SRCAP complex to DNA is suppressed by DNA methylation. Altogether, we propose that the SRCAP complex is the major determinant for preferential H2A.Z enrichment on unmethylated DNA, whereas DNA methylation destabilizes DNA wrapping in H2A.Z-containing nucleosomes.

Introduction

Structural features of the nucleosome, a protein-DNA complex composed of ~146 bp DNA wrapped around a histone octamer, directly affect and regulate genomic accessibility to a diversity of DNA binding proteins, in part, through incorporating histone variants with unique physical properties (Kurumizaka et al., 2021 [DOI](#); Talbert and Henikoff, 2016 [DOI](#)). The highly evolutionarily conserved H2A variant, H2A.Z, is involved in transcriptional regulation with a well described localization at the transcription start sites (TSSs) of active genes (Bönisch and Hake, 2012 [DOI](#); Giaimo et al., 2019 [DOI](#); Zlatanova and Thakar, 2008 [DOI](#)). However, while typically associated with gene activation, H2A.Z has also been connected to a range of functions including DNA repair, chromosome segregation, heterochromatin maintenance, and transcriptional repression (Colino-Sanguino et al., 2022 [DOI](#); Giaimo et al., 2019 [DOI](#)). Loss of H2A.Z induces lethality in many eukaryotes, such as mice, *Drosophila*, and *Tetrahymena* (Faast et al., 2001 [DOI](#); Liu et al., 1996 [DOI](#); Van Daal and

Elgin, 1992 [\[1\]](#)), and dysregulation of its localization has been linked to several diseases, including various cancers and the developmental disorder Floating-Harbor syndrome (Buschbeck and Hake, 2017 [\[2\]](#); Diegmüller et al., 2025 [\[3\]](#)). What mechanisms mediate proper H2A.Z localization, and how that connects with its broad-reaching functions, represents an important area of study.

Genomic sequencing analysis of a wide range of eukaryotes have found H2A.Z and DNA methylation, typically occurring as 5-methylcytosine (5mC), to be strikingly anti-correlated across the genome, at both a global and local level (Berta et al., 2021 [\[4\]](#); Coleman-Derr and Zilberman, 2012 [\[5\]](#); Conerly et al., 2010 [\[6\]](#); Murphy et al., 2018 [\[7\]](#); Nie et al., 2019 [\[8\]](#); Zemach et al., 2010 [\[9\]](#); Zilberman et al., 2008 [\[10\]](#)). The high conservation of this antagonism points to the existence of a shared fundamental pathway involved in creating this dichotomy and emphasizes its functional importance. In *Arabidopsis*, H2A.Z mediates recruitment of the demethylase ROS1, thereby promoting DNA demethylation; however, this pathway uses plant-specific machinery that targets only a particular set of genomic loci (Nie et al., 2019 [\[8\]](#)). More general mechanisms for how the mutually exclusive H2A.Z and DNA methylation compartments exist at large remain to be resolved.

DNA methylation may mediate H2A.Z exclusion via two distinct mechanisms: 1) influencing the intrinsic physical stability of the nucleosome, and/or 2) modulating the activity of nucleosome-remodelers. H2A.Z and H2A diverge in sequence along areas important for stabilizing intra-nucleosome interactions, such as the L1 loop, docking domain, and C-terminus (Bönisch and Hake, 2012 [\[11\]](#)). DNA methylation can influence the geometry of DNA, particularly its roll and twist, which then likely affect nucleosome wrapping (Li et al., 2022b [\[12\]](#)). A large body of work has been conducted on the physical impacts of either H2A.Z or DNA methylation on the nucleosome, but conflicting conclusions have been reported; incorporation of H2A.Z or DNA methylation independently has been shown to cause stabilizing (Chen et al., 2013 [\[13\]](#); Choy et al., 2010 [\[14\]](#); Collings et al., 2013 [\[15\]](#); Dai et al., 2021 [\[16\]](#); Ishibashi et al., 2009 [\[17\]](#); Lee et al., 2015 [\[18\]](#); Lee and Lee, 2012 [\[19\]](#); Park et al., 2004 [\[20\]](#); Thambirajah et al., 2006 [\[21\]](#)), destabilizing (Horikoshi et al., 2019 [\[22\]](#); Jimenez-Useche et al., 2014 [\[23\]](#); Jimenez-Useche and Yuan, 2012 [\[24\]](#); Lee et al., 2015 [\[18\]](#); Li et al., 2023 [\[25\]](#); Ngo et al., 2016 [\[26\]](#); Rudnizky et al., 2016 [\[27\]](#); Weber et al., 2010 [\[28\]](#); Zhang et al., 2005 [\[29\]](#)), or no detectable effects (Fujii et al., 2016 [\[30\]](#); Langecker et al., 2015 [\[31\]](#); Osakabe et al., 2015 [\[32\]](#)). Nevertheless, it is possible that the combination of H2A.Z and DNA methylation's effects may physically bias against retention of H2A.Z nucleosomes on methylated DNA and contribute to their overall antagonism.

H2A.Z remodeling is primarily carried out by the Ino80 family of nucleosome remodelers. Both the SRCAP complex (SRCAP-C) and Tip60 complex (Tip60-C) mediate exchange of canonical H2A to H2A.Z within a nucleosome (Ruhl et al., 2006 [\[33\]](#)), while the Ino80 complex (Ino80-C) has been reported to evict H2A.Z (Scacchetti and Becker, 2021 [\[34\]](#); Watanabe and Peterson, 2010 [\[35\]](#)). Tip60-C can also associate with the H2A.Z chaperone, ANP32e, to evict H2A.Z from nucleosomes (Mao et al., 2014 [\[36\]](#); Obri et al., 2014 [\[37\]](#)). Members of this remodeler family are prime candidates for enforcing the separation of H2A.Z and DNA methylation; however, it is unknown whether any of these complexes exhibit DNA methylation sensitivity.

To determine molecular mechanisms contributing to the mutual exclusion of H2A.Z and DNA methylation within the genome, here we investigated the effects of DNA methylation on both the physical stability of H2A.Z nucleosomes and remodeler-mediated deposition of H2A.Z onto synthetic DNA constructs by employing cryo-EM structural analysis and the physiological *Xenopus* egg extract system, respectively.

Results

Cryo-EM analysis of H2A.Z nucleosomes in the presence or absence of DNA methylation

To assess the physical impacts of DNA methylation on H2A.Z nucleosomes, we solved cryo-EM structures of human H2A.Z nucleosomes reconstituted with either methylated (Me) or unmethylated (UM) Sat2R-P DNA (Fig 1 [↗](#), Suppl Fig 1 [↗](#), Suppl Table 1 [↗](#)). The Sat2R-P sequence is a 152 bp artificial palindromic sequence containing 9 evenly distributed CpG sites and was engineered from a published canonical nucleosome crystal structure containing a methylated derivative of the human satellite II sequence (termed Sat2R) (Osakabe et al., 2015 [↗](#)) (Fig 1A [↗](#)). This sequence was chosen for two reasons: 1) human satellite II (HSat2) represents a highly abundant methylated sequence which undergoes disease-associated differential methylation (Ehrlich, 2003 [↗](#); Jackson et al., 2004 [↗](#); Tilman et al., 2012 [↗](#); Unoki, 2021 [↗](#); Wong et al., 2001 [↗](#)), and 2) the symmetric palindromic sequence was needed to increase final structure resolutions due to the two-fold pseudo symmetry about the nucleosome dyad axis. Prepared nucleosomes were run on a native PAGE gel to assess for quality, where both samples appeared as a singular band with minor comparable populations of unincorporated DNA (Suppl Fig 1A [↗](#)).

Cryo-EM structures of UM and Me Sat2R-P H2A.Z nucleosomes were solved at a final resolution of 3.05 Å and 2.78 Å, respectively (Fig 1B and C [↗](#), Suppl Fig 1 [↗](#)). We found that the nucleosome dyad was shifted roughly two base pairs from the center of the Sat2R-P palindrome, introducing a sequence asymmetry despite our intended use of the palindromic sequence. The shifted nucleotide positions were determined using the higher resolution densities surrounding the nucleosome dyad, which allowed for pyrimidine and purine distinction (Suppl Fig 2A-D [↗](#)). Two atomic models (termed v1 and v2) with mirrored DNA sequences were generated for each structure to account for the resultant sequence asymmetry, with the final EM densities likely reflecting an average of these two models (Suppl Fig 2E [↗](#)).

As a whole, both the methylated and unmethylated structures resembled previously published H2A.Z nucleosome structures and exhibited a linker DNA asymmetry where one side of the nucleosome (distal face 2, DF2) contained a highly flexible linker DNA region compared to the other (distal face 1, DF1) (Fig 1B and C [↗](#)) (Horikoshi et al., 2013 [↗](#); Lewis et al., 2021 [↗](#); Suto et al., 2000 [↗](#)). As a palindromic DNA sequence was used, it is likely that the asymmetrical faces in the cryo-EM structures arose due to an averaging of closed and open H2A.Z nucleosome particles into the same structure rather than representing an asymmetry found in individual particles.

In a direct comparison of DF2 linker DNA lengths, we found that the Me Sat2R-P map had much weaker linker DNA density than UM Sat2R-P, resulting in a ~7 bp shorter resolved DNA model for the methylated structure (Fig 1D [↗](#)). This loss of linker density indicates destabilized histone-DNA interactions and increased DNA structural variability in the presence of DNA methylation. Along similar lines, Root Mean Squared Distance (RMSD) analysis identified the H2A.Z-H2B dimer of DF2, as well as the overall DNA backbone, as regions of considerable difference between the methylated and unmethylated structures, whereas DF1, the closed side of the nucleosome with a highly resolved linker DNA, was largely similar between the two (Fig 1E [↗](#)).

DNA methylation creates open H2AZ nucleosome structures

To further characterize areas of change, maps of unmethylated and methylated H2A.Z structures were filtered and resampled to the same resolution (~3 Å) for fair comparison (Fig 2A [↗](#)). Three major differences between the maps were identified. First, cryo-EM densities of the H3 N-terminus and H2A.Z C-terminus on DF2 were much weaker in the methylated structure as compared to unmethylated, preventing model construction in these areas (Fig 2B [↗](#)). Second, an extra density on the H2A.Z-H2B acidic patch was seen on the methylated map but was not present on the unmethylated map (Fig 2C [↗](#)). Third, between the two major orientations the H4 N-terminal tail

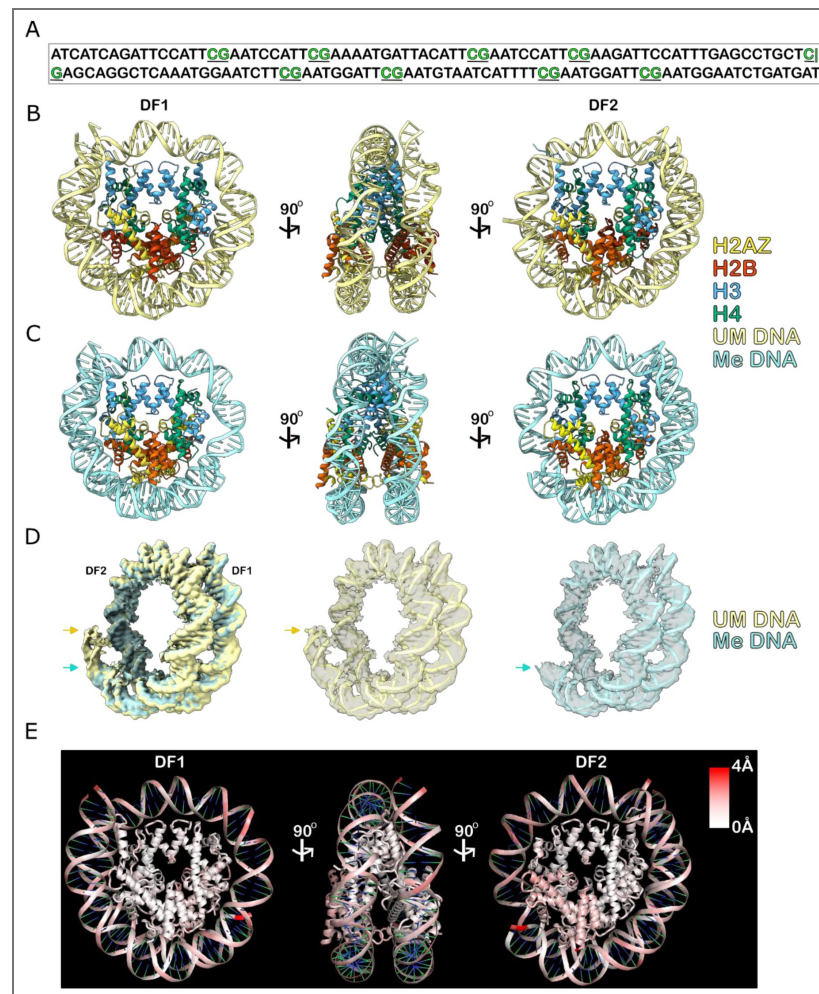


Figure 1. Cryo-EM structures of methylated and unmethylated Sat2R-P human H2A.Z nucleosomes.

(A) Diagram of the palindromic Sat2R-P sequence. Methylated cytosines are labeled in green and underlined. Vertical bar denotes halfway point. **(B and C)** Atomic models of unmethylated **(B)** and methylated **(C)** Sat2R-P H2A.Z structures (the v1 model for both). Two face views (DF1 and DF2) and a side view (middle) are shown. **(D)** Left, overlay of unmethylated (UM) and methylated (Me) Sat2R-P DNA EM densities. Middle and right, overlays of DNA densities on top of the corresponding v1 DNA atomic models for either the unmethylated (middle) or methylated (right) structures. Arrows point to the visualizable end of the DF2 linker DNA for each structure. **(E)** RMSD analysis comparing differences between unmethylated and methylated v1 Sat2R-P H2A.Z atomic models.

can adopt (Arimura et al., 2021 [DOI](#)), the unmethylated map showed a strong preference for the inward orientation whereas the methylated structure exhibited similar populations of inward and outward orientations (Fig 2D [DOI](#)).

The H3 N-terminus interacts with the H2A/H2A.Z C-terminus and DNA to hold the linker DNA close to the body of the nucleosome (Li et al., 2023 [DOI](#); Li and Kono, 2016 [DOI](#)). Loss of the H3 N-terminal density from the methylated structure suggests destabilized DNA-histone interactions that are consistent with the shorter resolved linker DNA on DF2 of the methylated model compared to unmethylated (Fig 1D [DOI](#)). The H4 N-terminal tail, on the other hand, mediates inter-nucleosomal interactions through contacts with a neighboring acidic patch (Chen et al., 2017 [DOI](#); Luger et al., 1997 [DOI](#)). The extra density on the Me Sat2R-P acidic patch may then have come from the H4 tail of another nucleosome, in line with the changes also observed in H4 tail orientation. These observations may be consistent with reports that DNA methylation increases condensation of nucleosomes, as well as bare DNA, *in vitro* and *in vivo* (Buitrago et al., 2021 [DOI](#); Jimenez-Useche et al., 2014 [DOI](#); Yang et al., 2020 [DOI](#)).

To more quantitatively assess methylation-dependent structural changes, we conducted an *in silico* mixing 3D classification analysis (Arimura et al., 2021 [DOI](#); Hite and MacKinnon, 2017 [DOI](#)) by combining micrographs collected from both methylated and unmethylated nucleosome samples and then sorted particles picked from the combined micrographs into six 3D classes of nucleosomes, each with varying degrees of open or closed linker DNA (Fig 2E and F [DOI](#), Suppl Fig 3 [DOI](#)). Most of the unmethylated particles (>60%) were sorted into the closed linker classes (Class 1 and 2) whereas the majority of methylated particles (~60%) belonged to classes with open linkers (Classes 3, 4, 6) (Fig 2F [DOI](#)). The fraction of hexasomes (Class 5) is minor in both conditions with a slight increase in methylated particles (Class 5). Among these open linker classes that are preferentially found in methylated particles, class 4 also represents nucleosomes where the palindromic center of the DNA has slid and is considerably offset from the dyad axis of the histone octamer. This may explain the weaker DF2 linker density seen in the high resolution Me Sat2R-P structure (Fig 1D [DOI](#)). Additionally, the DNA backbone of the methylated nucleosome structure had a lower local resolution compared to unmethylated, despite the methylated map having a higher global resolution (Suppl Fig 4 [DOI](#)). This is consistent with the increased population of open linker and slid nucleosome particles in the methylated structure. Altogether, our cryo-EM structural analysis suggests that the presence of DNA methylation creates open and potentially destabilized H2A.Z nucleosome structures.

Cryo-EM structures of 601L H2A.Z nucleosomes show no difference between methylation status

In addition to the Sat2R-P nucleosomes, we also solved cryo-EM structures of human H2A.Z nucleosomes reconstituted on methylated and unmethylated 601L DNA, a palindromic version of the lefthand portion of the classic Widom 601 sequence (Suppl Fig 5 [DOI](#) and 6 [DOI](#), Suppl Table 1 [DOI](#)) (Lowary and Widom, 1998 [DOI](#)). Final maps of the unmethylated and methylated structures were obtained at 3.09 Å and 3.25 Å, respectively (Suppl Fig 5B-E [DOI](#)). RMSD analysis of the solved structures showed that both maps were essentially identical to each other, with the few differences present localized to highly flexible regions of the nucleosome (Suppl Fig 5F [DOI](#)). The lack of clear structural differences when using 601L DNA is likely due to its artificially strong nucleosome positioning ability, which may have prevented detection of subtle changes caused by DNA methylation.

DNA methylation increases accessibility of H2A.Z but not H2A nucleosomes

In order to validate the observed opening of the H2A.Z nucleosome structure by DNA methylation, and to determine if this methylation effect is also present on nucleosomes containing canonical H2A, we measured nucleosome DNA accessibility via restriction enzyme digest (Fig 3 [DOI](#), Suppl Fig 7 [DOI](#) and 8 [DOI](#)). We tested both human H2A and H2A.Z nucleosomes reconstituted with methylated

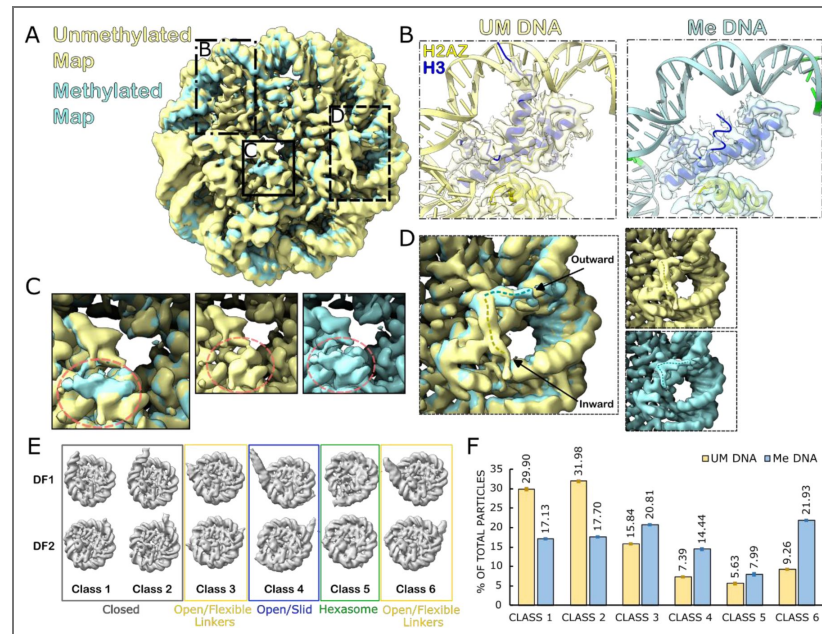


Figure 2. DNA methylation reduces H3-DNA contacts, alters H4 N-terminal tail orientations, and forms open nucleosome structures.

(A) Density maps of unmethylated (UM) and methylated (Me) Sat2R-P H2A.Z nucleosomes filtered and resampled to $\sim 3\text{\AA}$. The face shown, and all following zoom-ins, is of DF2. Maps are overlaid onto each other. Boxed regions highlight major areas of difference shown in B-D. (B) Zoom-in of the H3 N-terminus on DF2 of both structures. Density maps are shown as a transparent overlay on top of the corresponding v1 atomic models. (C) Zoom-in of the DF2 acidic patch on both structures. Area corresponding to the extra density found in the Me structure labeled in red. (D) Zoom-in of the H4 N-terminal tail. Dotted lines show each structure's preferred orientation (E) 3D Models used in the *in silico* mixing 3D classification analysis. Classes are as follows: (1 and 2) closed nucleosome structures with varying linker lengths; (3 and 6) open nucleosomes with highly flexible linkers; (4) open nucleosomes that have shifted position from center; (5) hexasomes. Both distal faces (DF1 and DF2) are shown of each model. (F) Results of classification analysis. Y-axis is the proportion of particles sorted into each class as a percent of the total input particles from each sample (UM or Me). Error bars represent SEM, $n = 3$ technical replicates.

and unmethylated versions of 152 bp non-palindromic Sat2R DNA that was modified to include a singular *HinfI* site 18 bp in from the 5'-end containing a Cy5-fluorophore (Fig 3A [↗](#)). Homogeneity of the reconstituted nucleosomes was assessed via native PAGE and mass photometry analysis prior to restriction digest analysis (Suppl Fig 7A-C [↗](#)). Digest of bare control DNA confirmed that *HinfI* behaved similarly on both methylated and unmethylated Sat2R substrates over the time scale used in the experiment (Suppl Fig 7D [↗](#)). We found that H2A.Z nucleosomes were more accessible to enzymatic digest than H2A nucleosomes, independent of DNA methylation status (Fig 3B and C [↗](#), Suppl Fig 8 [↗](#)). When comparing methylation-dependent differences for each type of nucleosome, methylated and unmethylated H2A nucleosomes were largely digested at the same rate. Methylated H2A.Z nucleosomes, on the other hand, exhibited a small but more pronounced increase in accessibility as compared to its unmethylated counterpart (Fig 3B and C [↗](#), Suppl Fig 8 [↗](#)). These results corroborate our structural findings and also suggest that the effects of DNA methylation may be negligible on an intrinsically stable nucleosome (i.e. containing H2A) but can further destabilize, in a selective manner, the already open and accessible H2A.Z nucleosome.

H2A.Z and DNA methylation localization in *Xenopus laevis* sperm pronuclei and a fibroblast cell line

Results from our structural and biochemical experiments indicate decreased wrapping stability of H2A.Z nucleosomes on methylated DNA. While functional implications likely exist, this subtle effect on the intrinsic physical stability of the H2A.Z nucleosome is unlikely to be the sole driver of the mutually exclusive H2A.Z and DNA methylation compartmentalization in cells. To identify the major biochemical mechanism underlying this dichotomy, we turned to the *Xenopus laevis* egg extract system, a unique cell-free system that allows for characterization of chromatin proteins under physiological conditions (Jenness et al., 2018 [↗](#); Lohka and Maller, 1985 [↗](#); Postow et al., 2008 [↗](#); Wassing et al., 2024 [↗](#); Xue et al., 2013 [↗](#); Zierhut et al., 2014 [↗](#)).

To ensure that *Xenopus* egg extract is a valid tool for studying the H2A.Z-DNA methylation antagonism, we conducted CUT&Tag – Bisulfite Sequencing (CnT-BS) (Li et al., 2021 [↗](#)), a method which allows for the direct assessment of DNA methylation on CUT&Tag reads, against H2A.Z of *Xenopus* sperm nuclei incubated in interphase extract. This condition mimics fertilization and allows for full nucleosome assembly onto sperm (primarily composed of protamines and H3/H4 tetramers) to create functional pronuclei (Oikawa et al., 2020 [↗](#)). We also collected samples against H3 to serve as a pan-nucleosome control. Since sperm pronuclei in egg extract represents a unique transcriptionally silent state (Engelke et al., 1980 [↗](#); Newport and Kirschner, 1982 [↗](#); Veenstra et al., 1999 [↗](#)), we also performed CnT-BS on XTC-2 cells, a *Xenopus laevis* fibroblast cell line (Pudney et al., 1973 [↗](#)), to assess changes in DNA methylation and H2A.Z overlap across different nuclei types.

Initial comparisons of CnT-BS reads across two biological replicates of our four conditions demonstrated good reproducibility between replicates, while each set of experimental conditions exhibited distinct characteristics. A greater than 99 % bisulfite conversion efficiency rate, as assessed through our unmethylated lambda DNA spike-in, was achieved, and correlation matrices of both the mapped reads and methylation calls were highly similar between biological replicates (Suppl Fig 9 [↗](#), Suppl Table 2 [↗](#) and 3 [↗](#)). When comparing across conditions, for both the H3/H2A.Z mapped reads and DNA methylation sites, the first degree of difference seen was between XTC-2 and sperm samples, and then H3 vs H2A.Z following after, indicating that large-scale changes in the DNA methylation and genomic landscape occurred between the two different nuclear states (Suppl Fig 9 [↗](#)).

To assess characteristics of H2A.Z and H3 genomic distributions, we subset the *Xenopus* genome into 1,000 bp bins and classified the top 20 % of bins containing the most mapped reads for each condition based on annotated functionalities (Fisher et al., 2023 [↗](#)). We found that a greater proportion of highly mapped H2A.Z bins were located near promoter regions as compared to H3 for both sperm and XTC-2 samples (Suppl Fig 10A [↗](#)). Although transcription is suppressed in *Xenopus* egg extracts, H2A.Z in sperm pronuclei were well positioned at TSSs, as generally seen for actively transcribed genes in somatic cells (Suppl Fig 10B and C [↗](#)). This mirrors the reported

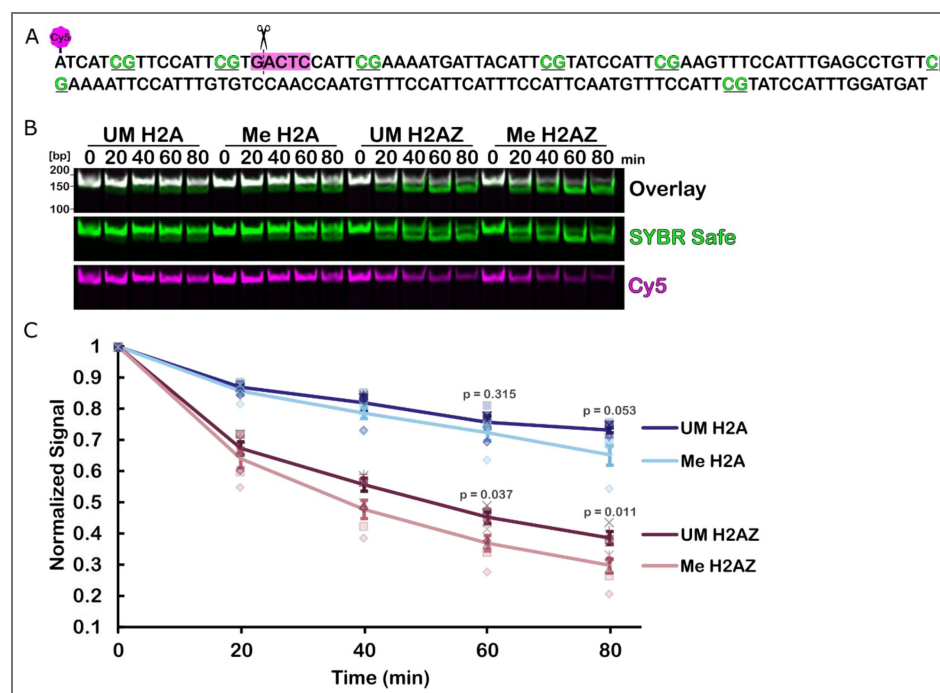


Figure 3. DNA methylation increases H2A.Z nucleosome accessibility.

(A) Diagram of the 1HinfI_Sat2R DNA sequence used. CpG sites are shown in green and underlined. HinfI recognition site highlighted in magenta. A Cy5 fluorophore is attached to the 5' end nearest the HinfI site. **(B)** Native gel showing HinfI digestion time course of UM and Me human H2A and H2A.Z nucleosomes. SYBR Safe was used to stain total DNA. Digestion efficiency was assessed via loss of Cy5 signal and a downward shift in the total DNA band. **(C)** Quantification of **(B)**. Cy5 signals were normalized to total SYBR Safe signal for each sample. Error bars represent SEM. Each shape represents data from one independent experiment, $n = 5$ experiments. Statistics comparing unmethylated and methylated samples at indicated timepoints were completed with a two-tailed t-test assuming unequal variance with p -values listed above each comparison.

deposition of H3K4me3 at the +1 TSS sites in *Xenopus* sperm nuclei prior to remodeling as well as reports in zebrafish sperm and *Drosophila* that H2A.Z is positioned at promoters prior to zygotic genome activation (ZGA) (Ibarra-Morales et al., 2021 [DOI](#); Murphy et al., 2018 [DOI](#); Oikawa et al., 2020 [DOI](#)).

H2A.Z is biased towards hypomethylated regions of the genome in both *Xenopus* sperm pronuclei and the XTC-2 fibroblast cell line

We next determined the methylation status of CpGs associated with either H2A.Z or H3 across replicates of XTC-2 and sperm nuclei (Fig 4A [DOI](#)). One notable drawback of CUT&Tag is a bias towards open and accessible regions (Kaya-Okur et al., 2020 [DOI](#); Richard et al., 2025 [DOI](#)), which may artificially amplify open chromatin signals and lead to an underestimation of DNA methylation status. To circumvent this issue, we analyzed H3 CpG methylation in areas outside of H2A.Z peaks (which tend to be highly accessible) to compare genomic methylation profiles in H2A.Z low/absent regions (H3) vs H2A.Z enriched peaks (H2A.Z).

Overall, sperm chromatin had a higher degree of methylation than XTC-2 chromatin, with 85.3 ± 0.4 % s.d. of CpGs in H3 reads experiencing methylation in sperm compared to 53.5 ± 0.4 % in XTC nuclei (Suppl Table 2 [DOI](#)). Methylated CpG frequencies of H2A.Z-associated reads were reduced to 43.0 ± 0.7 % in sperm chromatin and even further down to 3.0 ± 0.3 % in XTC-2 cells. Mapping of H3 associated methylation calls to annotated genes of the *Xenopus laevis* genome showed ubiquitous distribution of methylation across the length of genes, except for at the TSSs where there was a marked depletion of methylation (Suppl Fig 10B and C left [DOI](#)). Nearly all of these H3-marked hypomethylated TSSs exhibited H2A.Z enrichment in both sperm pronuclei and XTC-2 (Suppl Fig 10B and C right [DOI](#)), indicating that hypomethylation-exclusive positioning of H2A.Z is most prominent at the TSSs (Suppl Fig 10D [DOI](#)).

DNA methylation-sensitive H2A.Z deposition on synthetic DNA constructs in *Xenopus* egg extract

To directly test whether the presence or absence of DNA methylation can dictate preferential H2A.Z deposition independently of other pre-existing epigenetic marks or DNA sequence, we took advantage of the *Xenopus* egg extract's ability to efficiently assemble nucleosomes *de novo* onto naked DNA templates (Laskey et al., 1977 [DOI](#)) (Fig 4B [DOI](#)). We found that, upon incubation of DNA coated magnetic beads in interphase egg extract, H2A.Z accumulated less efficiently on methylated DNA substrates compared to unmethylated (Fig 4C and D [DOI](#), Suppl Fig 11 [DOI](#)), whereas this methylation-sensitive deposition was not seen for H2A.X-F, the predominant H2A histone variant in *Xenopus* egg extract (Fig 4E and F [DOI](#), Suppl Fig 11 [DOI](#)) (Shechter et al., 2009 [DOI](#)). The methylation bias of H2A.Z held true for arrays of either the GC-rich Widom 601 (57 % GC content) or the more AT-rich human HSat2 (38% GC content) sequence. We next asked whether both H2A.Z variants, H2A.Z.1 and H2A.Z.2 (H2A.V), exhibit the same DNA methylation sensitivity. Since our antibody does not distinguish between the two H2A.Z paralogs, we translated ALFA-tagged (Götzke et al., 2019 [DOI](#)) versions of the variants in egg extract from exogenously added mRNA and found that both paralogs displayed preferential deposition onto unmethylated DNA (Suppl Fig 12 [DOI](#)).

The SRCAP complex mediates H2A.Z's preferential deposition on unmethylated DNA

Since SRCAP-C is known to play a major role in depositing H2A.Z to nucleosomes via exchange of H2A dimers on the nucleosome with H2A.Z (Ruhl et al., 2006 [DOI](#)), we examined if the observed preferential association of H2A.Z with unmethylated DNA in *Xenopus* egg extracts is mediated by SRCAP-C. Depletion of SRCAP-C from extract using a custom antibody against *Xenopus laevis* SRCAP greatly reduced the amount of H2A.Z associated with unmethylated DNA compared to control depletion (Δ IgG), while its levels on methylated DNA were not affected (Fig 5A-C [DOI](#), Suppl Fig 13A [DOI](#)). Consequently, in SRCAP-C-depleted (Δ SRCAP) extracts, residual H2A.Z binding to DNA became insensitive to DNA methylation status, while the amount of H2A.X-F on DNA was not

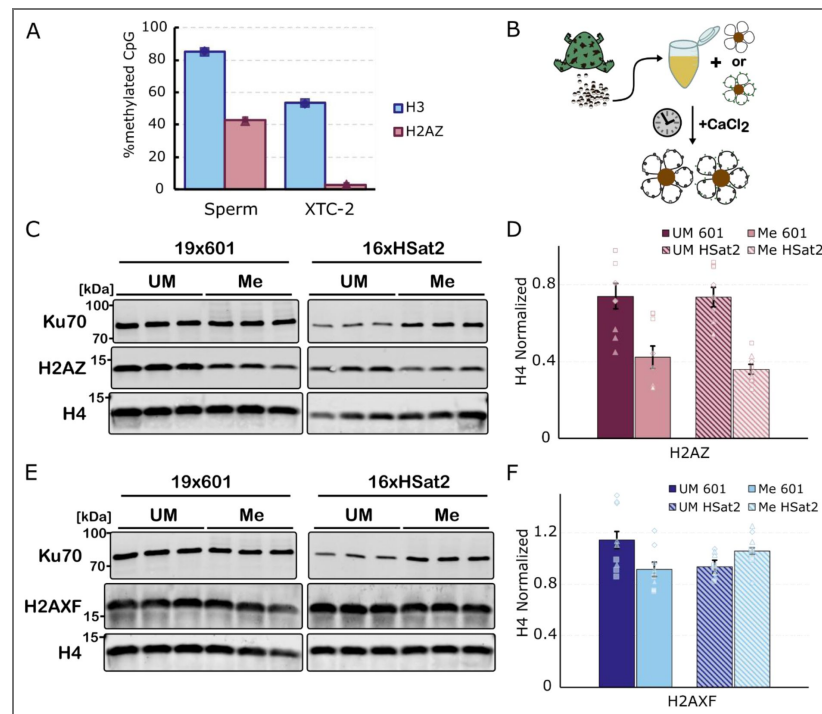


Figure 4. Presence or absence of DNA methylation influences H2A.Z deposition.

(A) Percentage of methylated CpG sites ($\geq 5\times$ coverage) associated with H2A.Z peaks or H3 reads in sperm pronuclei incubated in interphase egg extract or XTC-2 nuclei following CnT-BS library preparation. H3 data is taken from regions outside of H2A.Z enriched peaks. Data points represent biological replicates ($n = 2$). **(B)** Schematic of chromatinization assay. Magnetic beads coated with DNA of interest were incubated in *Xenopus* egg extract with the addition of CaCl_2 to induce cycling into interphase. After 60 min, DNA beads were isolated to assess for histone composition. **(C and E)** Western blot results of chromatinization assay probing for either H2A.Z **(C)** or H2A.X-F **(E)**. Ku70 and H4 signals are shown as loading controls. Beads coated with unmethylated (UM) or methylated (Me) 19-mer arrays of Widom 601 (19x601) or 16-mer arrays of HSat2 (16xHSat2) were used. A representative blot of technical triplicates for each condition is presented. **(D and F)** Quantification of **(C and E)**. H2A.Z and H2A.X-F signals were normalized to H4 intensity. Error bars represent SEM. Data points represent technical triplicates across 3 biological replicates with each shape representing data from a single independent experiment ($n = 9$).

affected by SRCAP-C depletion (Fig 5B and D [↗](#), Suppl Fig 13A [↗](#)). Consistent with the critical role of SRCAP-C in the preferential deposition of H2A.Z to unmethylated DNA, DNA-binding of SRCAP and ZNHIT1, two core subunits of SRCAP-C, was greatly inhibited by methylation in egg extract (Fig 5E-F [↗](#), Suppl Fig 13B [↗](#)). In contrast, p400, the ATPase of Tip60-C, did not exhibit methylation-sensitive DNA binding (Suppl Fig 14 [↗](#)). From these results, we conclude that SRCAP-C is the major driver biasing H2A.Z deposition against methylated DNA.

Discussion

In this study, we found that DNA methylation can affect H2A.Z deposition and stability via two distinct mechanisms (Fig 6 [↗](#)). DNA methylation directly opposes H2A.Z deposition by inhibiting SRCAP-C DNA binding. SRCAP-C accounts for the majority of H2A.Z deposition on unmethylated DNA and is the primary mediator behind DNA methylation's exclusion of H2A.Z in the transcriptionally inactive *Xenopus* egg extract. A subpopulation of H2A.Z, however, is still deposited onto methylated DNA, driven by methylation insensitive mechanism(s) whose identity remains to be established. Once formed, H2A.Z nucleosomes on methylated DNA are more open and accessible compared to their unmethylated counterparts. This decrease in DNA-histone contact stability of methylated H2A.Z nucleosomes may then further facilitate clearance of H2A.Z from methylated regions of the genome.

Supporting our observation that a SRCAP-independent mechanism(s) exists to deposit H2A.Z on methylated DNA in *Xenopus* egg extract, a substantial fraction (~40%) of H2A.Z-associated genomic DNA is methylated in sperm pronuclei. In the fibroblast cell line XTC-2, that methylated H2A.Z fraction is greatly reduced to 3%. However, due to the accessibility bias inherent within CUT&Tag, our data likely underestimates the true degree of H2A.Z and DNA methylation overlap, particularly in the XTC-2 samples. For comparison, a previous study in human myometrium reported that ~20% of CpGs overlapping with H2A.Z are methylated, indicating that a sizeable population of colocalized H2A.Z and DNA methylation exists in somatic cells which warrants further investigation (Berta et al., 2021 [↗](#)). Nevertheless, we believe the decreased trend in H2A.Z and DNA methylation overlap of XTC-2 cells holds true and suggests that the high degree of H2A.Z and DNA methylation coincidence in sperm pronuclei reflects its unique early development and/or transcriptionally silent state. Given the strong preference of SRCAP-C binding to unmethylated DNA in *Xenopus* egg extracts, we predict that additional pathways missing in the egg extract system are necessary to reinforce the antagonism between DNA methylation and H2A.Z enrichment.

Perhaps the greatest difference between XTC-2 cells and *Xenopus* eggs is the presence of transcription, which is one of the main evictors of H2A.Z and necessary for global H2A.Z turnover as well as pruning of randomly deposited H2A.Z within gene bodies (Hardy et al., 2009 [↗](#); Lashgari et al., 2017 [↗](#); Ranjan et al., 2020 [↗](#); Tramantano et al., 2016 [↗](#)). Traversing of RNA polymerase (Pol) II across nucleosomes, aided by the FACT complex, results in transient histone dimer displacement (Belotserkovskaya et al., 2003 [↗](#); Bevington and Boyes, 2013 [↗](#); Bintu et al., 2011 [↗](#); Kireeva et al., 2002 [↗](#)), leading to the possibility that the more accessible H2A.Z nucleosome structures formed on methylated DNA may have increased susceptibility to Pol II disruption.

Furthermore, FACT and Spt6 do not reincorporate H2A.Z into the reassembled nucleosome post Pol II encounter, reinforcing selective loss of H2A.Z (Heo et al., 2008 [↗](#); Jeronimo et al., 2015 [↗](#)).

Outside of transcription, the egg extract system also lacks both the DNMT3 enzymes responsible for *de novo* methylation and the TET system responsible for DNA methylation removal in animals (Wühr et al., 2014b [↗](#)). In *Arabidopsis*, H2A.Z promotes ROS1-associated DNA demethylation (Nie et al., 2019 [↗](#)) and must be excluded to ensure faithful heterochromatin formation (Zhou et al., 2023 [↗](#)). Thus, H2A.Z-driven mechanisms to limit DNA methylation remain relevant pathways to increase the H2A.Z/DNA methylation separation within the genome which may be missing in eggs/early embryonic states.

Figure 5. The SRCAP complex mediates H2A.Z's preferential association with unmethylated DNA.

(A) Western blot to detect SRCAP in control IgG-depleted *Xenopus* egg extract (Δ IgG) or SRCAP-depleted extract (Δ SRCAP). Bottom panel shows ponceau staining as a loading control. (B) Western blots of chromatinization assay in IgG control vs SRCAP depleted egg extract. Magnetic beads coated with 19-mer arrays of Widom 601 (19x601) or 16-mer arrays of HSat2 (16xHSat2) were incubated with interphase egg extract for 60 min, before their isolation. (C) Quantification of H2A.Z signal normalized to H4 from (B). (D) Quantification of H2A.X-F signal normalized to H4 from (B). Error bars in (C) and (D) represent SEM from $n = 3$ biological replicates and each shape represents data from one independent experiment. (E) Western blot staining for SRCAP on 16xHSat2 DNA beads incubated in interphase egg extract. (F) Western blot staining for ZNHIT1 on specified DNA beads incubated in interphase egg extract. Representative image shown from two independent experiments.

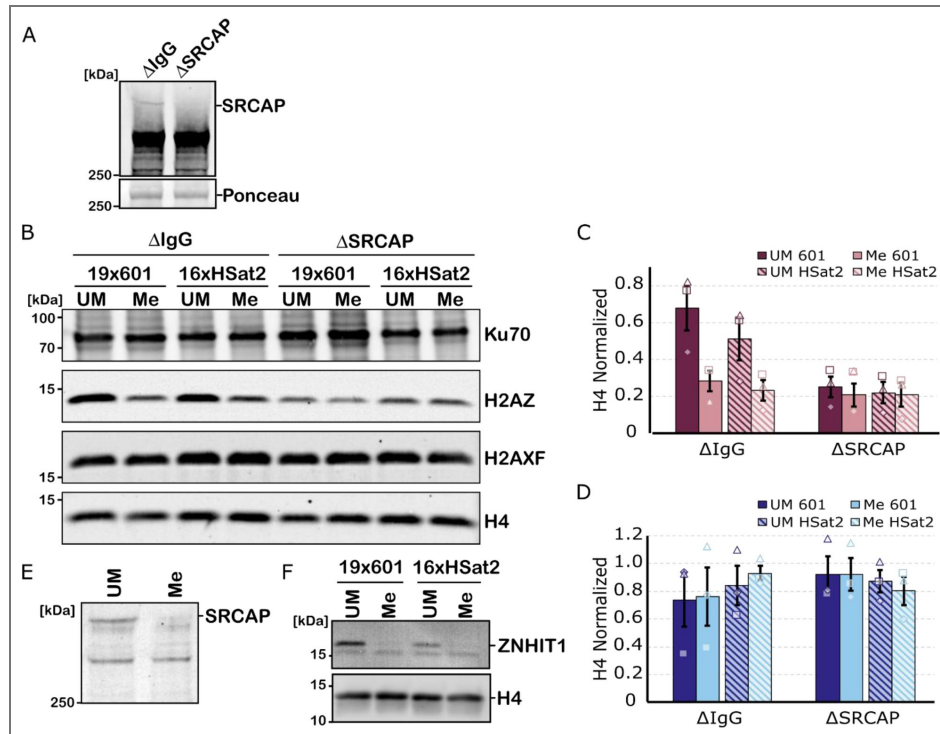
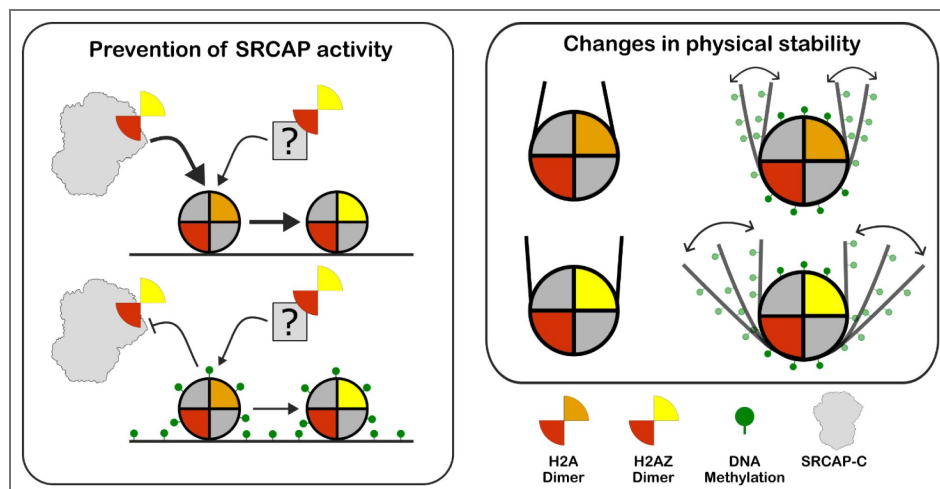


Figure 6. DNA methylation prevents SRCAP-C mediated H2A.Z deposition and creates more open and accessible H2A.Z nucleosome structures.

Left, SRCAP-C is capable of depositing H2A.Z on unmethylated substrates, but presence of DNA methylation prevents binding and deposition activity. Another unidentified mechanism deposits H2A.Z in a DNA methylation insensitive manner. Right, DNA methylation decreases DNA wrapping stability, an effect felt greatest on the already open H2A.Z nucleosomes compared to the more tightly wrapped H2A nucleosomes, creating more open and accessible H2A.Z nucleosome structures.



Exactly how SRCAP-C senses DNA methylation remains a pertinent question. Recent structural work on the yeast SWR1 complex and human SRCAP-C indicate that both the ATPase and the ZNHIT1 subunit (Swc6 in yeast) mediate binding of, and one-dimensional diffusion on, naked DNA substrates (Louder et al., 2024 [DOI](#); Park et al., 2024 [DOI](#)), marking them as likely candidates responsible for methylation sensing. Additionally, the YEATS domain-containing subunit GAS41, found in both SRCAP-C and Tip60-C, recognizes H3K14ac and H3K27ac (Hsu et al., 2018a [DOI](#), 2018b [DOI](#)), and deposition of H3K27ac, itself, is antagonized by DNA methylation at enhancers (King et al., 2016 [DOI](#)). It is possible that both direct recognition of DNA methylation and changes in the PTM landscape of methylated chromatin play a role in dictating SRCAP-C localization and activity.

Although a consensus on how DNA methylation influences nucleosome stability has yet to be achieved, our results indicate that DNA methylation creates more open nucleosome structures, particularly on H2A.Z nucleosomes, similar to reports from FRET-based and molecular simulation studies on the canonical H2A nucleosomes (Jimenez-Useche et al., 2014 [DOI](#); Jimenez-Useche and Yuan, 2012 [DOI](#); Li et al., 2022a [DOI](#); Ngo et al., 2016 [DOI](#)). Although the ambiguity in DNA positioning of our cryo-EM structures prevents us from discussing the structural coordination of specific CpG sites, we favor the idea that the increase in accessibility is due to effects of DNA methylation on global DNA structure rather than at specific methylated segments. This is in line with observations that methylation increases overall DNA stiffness, making it harder for the DNA to bend and wrap around histone octamers (Ngo et al., 2016 [DOI](#); Pérez et al., 2012 [DOI](#)).

Our restriction enzyme accessibility assay shows that H2A.Z nucleosomes are considerably more open and accessible compared to canonical H2A, regardless of methylation status. This difference is substantially larger than that induced by DNA methylation on H2A.Z, suggesting that the physical influence of DNA methylation on nucleosomes is overall subtle. This apparent subtlety, however, may be due to the internal placement of the cut site that we tested, and thus the effect of DNA methylation was preferably observed on an already destabilized nucleosome with H2A.Z. Similar findings have been reported on canonical H2A 601 nucleosomes, where the effects of DNA methylation manifest greatest on an open nucleosome sub-population compared to their closed counterparts (Jimenez-Useche and Yuan, 2012 [DOI](#)). This is likely why we saw no methylation-dependent structural differences with the 601L sequence (Suppl Fig 5 [DOI](#)), as its innate nucleosome wrapping and positioning ability is even stronger than the classic Widom 601. Overall, we suggest that the subtle nucleosome destabilization effect of DNA methylation is selectively pronounced on H2A.Z nucleosomes, which already exhibit an intrinsic instability compared to canonical H2A nucleosomes.

Beyond its interaction with DNA methylation, we show that H2A.Z promoter localization occurs immediately upon conditions mimicking fertilization in a transcription-independent manner. Unlike the almost complete loss of histones seen in mammalian sperm (Ward and Coffey, 1991 [DOI](#); Wykes and Krawetz, 2003 [DOI](#)), *Xenopus* sperm is packaged as a mixture of protamines and H3/H4 tetramers with very low levels of H2A/H2B dimers, which are then remodeled into full nucleosomes upon fusion with an egg (Katagiri and Ohsumi, 1994 [DOI](#); Mann et al., 1982 [DOI](#); Oikawa et al., 2020 [DOI](#)). In our egg extract system, which lacks the maternal genome, we expect the bulk of H2A.Z in sperm pronuclei to be newly deposited and find that H2A.Z is still specifically targeted to promoters either via underlying sequence motifs (Ibarra-Morales et al., 2021 [DOI](#); Murphy et al., 2018 [DOI](#)) or tetramer modifications maintained in sperm (Oikawa et al., 2020 [DOI](#)). This is similar to *Drosophila* where H2A.Z is recruited to TSSs prior to both Pol II binding and ZGA, marking genes for zygotic activation (Ibarra-Morales et al., 2021 [DOI](#)), indicating conservation of this initial transcription-independent H2A.Z positioning across both vertebrates and invertebrates.

Our work here provides a critical step in the pathways dictating genomic organization by uncovering the previously undescribed methylation sensitivity of SRCAP-C along with the destabilizing effects of DNA methylation on H2A.Z nucleosomes. Together, these findings provide molecular mechanisms driving the conserved antagonism between H2A.Z and DNA methylation in the genome. However, many questions remain to be answered, such as why the existence of both methylation sensitive and insensitive H2A.Z deposition pathways, the broader functional implications of the H2A.Z-DNA methylation antagonism as a whole, and mechanisms driving its

maintenance in species that experience periods of global demethylation. These questions represent important areas of future study for our understanding of epigenome dynamics throughout development.

Materials and methods

Histone purification and refolding

pET3 vectors containing sequences for human histones H2A, H2A.Z, H2B, H3.2, and H4 (gifts from Shixin Liu) were expressed in Rosetta (DE3) cells (Novagene, Cat# 70954). Cells were pelleted 4000 x g, 20 min, 4 °C and resuspended in wash/lysis buffer [50 mM Tris-HCl pH 7.5, 100 mM NaCl, 1 mM EDTA, 5 mM β -mercaptoethanol, 0.2 mM PMSF] + 1 % Triton-X-100 then sonicated. Inclusion pellets were spun 25,000 x g, 20 min, 4 °C, and incubated with DMSO at room temperature (RT) for 30 min. Pellets were then resuspended in unfolding buffer [20 mM Tris-HCl pH 7.5, 7 M guanidine-HCl, 10 mM DTT], sonicated until pellets were fully broken up, and re-pelleted at 25,000 x g, 20 min, RT. The supernatant was then collected and dialyzed in urea dialysis buffer overnight at 4 °C [10 mM Tris-HCl pH 7.5, 7 M Urea, 1 mM EDTA, 5 mM β -mercaptoethanol, 100 mM or 200 mM NaCl for H2A/H2B and H3/H4, respectively]. Samples were then pelleted at 25,000 x g, 20 min, RT and the supernatant passed through Q Sepharose Fast Flow resin (Cytiva, Cat# 17051010). The flow-through was then collected and added to SP Sepharose Fast Flow resin (Cytiva, Cat# 17072910), washed with urea dialysis buffer, and eluted with elution buffer [10 mM Tris-HCl pH 7.5, 7 M Urea, 1 mM EDTA, 600 mM NaCl, 5 mM β -mercaptoethanol].

Equimolar ratios of histones for either octamer, or dimer and tetramer preparation were then mixed and dialyzed into refolding buffer [10 mM Tris-HCl pH 7.5, 1 mM EDTA, 5 mM β -mercaptoethanol, 2 M or 500 mM NaCl for octamers or dimers/tetramers, respectively]. Refolded histones were then purified via FPLC (ÅKTA pure, Cytiva, Cat# SN2626712) (Superdex 200 10/300 GL increase column, Cytiva, Cat# 28990944).

Generation of DNA arrays

Generation of HSat2 arrays for use in chromatinization assays or arrays of Sat2R-P halves to prepare palindromic sequences for cryo-EM analysis was done as previously described in (Dyer et al., 2003 [DOI](#)). Briefly, DNA sequences of interest (see Table 1 [DOI](#)) containing one BglII site, one BamHI site, and two EcoRV sites surrounding the sequence were generated and purchased from Integrated DNA Technologies (IDT™) and cloned into the pUC18 vector just downstream of the XbaI site. The plasmid was then digested with either XbaI (NEB Cat# R0145) and BglII (NEB Cat# R0144) (to obtain the vector) or BglII and BamHI (NEB Cat# R3136) (to obtain the insert), and agarose gel purified. Vector and insert DNA were then ligated with T4 DNA ligase (NEB Cat# M0202). The digest and ligation steps were repeated until the desired repeat number was achieved.

Palindromic DNA preparation for Cryo-EM

Arrays of Sat2R-P halves were generated as described above. Plasmid containing an array of 601L halves (601L-145-8X) was a gift from Curt Davey (Addgene plasmid # 158572; <http://n2t.net/addgene:158572> [DOI](#); RRID:Addgene_158572 [DOI](#)) (Chua et al., 2012a [DOI](#)). Arrays of palindromic DNA halves for Cryo-EM analysis were then digested with EcoRV (NEB Cat# R3195) and purified via PEG precipitation (0.5 M NaCl, 8 % w/v PEG8000). Supernatant were collected and further purified via phenol-chloroform extraction. Insert ends were dephosphorylated with QuickCIP (NEB Cat# M0525) and phenol-chloroform extracted. Samples were then digested with either HinfI (NEB Cat# R0155) (601L DNA) or AvaI (NEB Cat# R0152) (Sat2R-P DNA), phenol-chloroform extracted, and FPLC purified in 1xTE buffer (Superdex 200 10/300 GL increase column, Cytiva, Cat# 28990944). DNA halves were then incubated with T4 DNA ligase (NEB Cat# M0202) per manufacturer's protocol until all halves were completely ligated.

Sequence Name	Sequence	Vector	Plasmid Name	Insert Location	Source	Assay(s) Used In	Additional Notes
Sat2R-P	ATCATCAGATTCCATTGGAATCCATTGAAAA TGATTACATTGGAATCCATTGGAATCCAT TTGAGCCTGCTCGAGCAGGCTCAATGGA TCTTGAATGGATTGGAATGATCATTTTCG AATGGATTGGAATGGAATCTGATGAT	pUC18	RMSP81	Between XbaI and BamHI sites	Generated and purchased from Integrated DNA Technologies (IDT™)	Cryo-EM	Modified from the Sat2R sequence in (Osakabe et al, 2015).
601L	ATCACAATCCCGGTGCCGAGGCCGCTCAAT TGGTCGTAGACAGCTCTAGCACCCTTAA CGCAGCTACGGAATCCGTCGTGCGTTAA GCGGTCTAGAGCTGTCTACACCAATTGA GCGGCTCGGCACCGGATTGTGAT	pUC57	601L-145-8X	EcoRV	(601L-145-8X) was a gift from Curt Davey (Addgene plasmid # 158572; http://n2t.net/addgene:158572 ; RRID:Addgene_158572)	Cryo-EM	
1HinfI_Sat2R	ATCATCGTTCCATTGCTGACTCCATTGAAA ATGATTACATTGCTATCCATTGGAATTTCCA TTTGAGCCTGTTGAAAATCCATTGTGTCC AACCAATGTTCCATTATTCATTCAATGTT TCCATTGCTATCCATTGGATGAT	pUC18	RMSP161	Between XbaI and BamHI sites	Generated and purchased from Integrated DNA Technologies (IDT™)	Restriction Digest Assay	DNA generated via PCR amplification using following primers: /5Cy5/ATCATCGTTCC ATTCGTGAC; ATCATCCAAATGGATA CGAATGG
HSat2 arrays (1 repeat shown)	CGTTTGATTCCATTGATGTTGATTCCATTGCA GTCCATTGCGATGATAATTCATTGATCTTT GCGATGATTCCATTCTTTCCATTGAGATGA TTCCATTGAGACCATTCGATGATTGCATTCA ATTCAATCGATGACGATTATCAATCCGTTT AATGATTCCATTGCAATTCGAATGATGATGAT CCATTGATCCATTGATGATGATTCATGC GATTCCATTGATGATGACTCCGATATCGGA TCTGATATC	pUC18	RMSP64	Between XbaI and BamHI sites	Generated and purchased from Integrated DNA Technologies (IDT™)	Chromatinization Assays	GenBank: X06199.1
601 arrays (1 repeat shown)	CTGGAGAATCCCGGTGCCGAGGCCGCTCA ATTGGTCGTAGCAAGCTCTAGCACCCTTAA ACGCACGTACGCGCTGTCCCGCGTTTAA ACCGCCAAGGGGATTACTCCCTAGTCTCCA GGCAGGTGTGAGATATATACCTGTGTCAT GTATTGAACAGCGACTCGGGTTATGTGATGG ACCTATACGCGGCCGCC	pUC18	pAS696	EcoRI/XbaI	Gift from Aaron Straight	Chromatinization Assays	
Oligos for Sequencing Library Preparation							
Sequence Name	Sequence				Source	Assay(s) Used In	Additional Notes
Tn5mC-Apt1	T/iMe-dC/GT/iMe-dC/GG/iMe-dC/AG/iMe-dC/GT/iMe-dC/AGATGTGATATAAGAGA/iMe-dC/AG				(Li et al, 2021); Generated and purchased from Integrated DNA Technologies (IDT™)	CnT-BS adapter oligo	Annealed with Tn5mC1.1-A1block to form adapter oligo
Tn5mC1.1-A1block	/5Phos/CTGTCTCTTATACA/3ddC/				(Li et al, 2021); Generated and purchased from Integrated DNA Technologies (IDT™)	CnT-BS adapter oligo	Annealed with Tn5mC-Apt1 to form adapter oligo
Tn5mC-RepIO1	/5Phos/iMe-dC/TGT/iMe-dC/T/iMe-dC/TTATA/iMe-dC/A/iMe-dC/AT/iMe-dC/T/iMe-dC/iMe-dC/GAG/iMe-dC/iMe-dC/iMe-dC/A/iMe-dC/GAGA/iMe-dC/3InvdT/				(Li et al, 2021); Generated and purchased from Integrated DNA Technologies (IDT™)	CnT-BS	Used during oligonucleotide replacement step

Table 1. DNA sequences/oligos used in the study.

Methylation of DNA substrates

DNA substrates were enzymatically methylated using the *M.SssI* CpG methyltransferase (NEB, Cat# M0226). Approximately 160 U of methyltransferase were used for 50 µg of DNA in 1x NEB 2 buffer with 500 µM SAM. Samples were incubated at 37 °C for at least one hour. Completion of methylation was checked using the methylation sensitive restriction enzyme BstUI (NEB, Cat# R0518). DNA was purified using phenol-chloroform extraction prior to use.

Nucleosome Reconstitution

DNA and either histone octamers or dimers and tetramers were mixed at a final target concentration of 1 µM in 1x TE buffer and 2 M NaCl. Exact ratios of DNA to histones were titrated for each reconstitution batch, starting at equimolar ratios. Samples loaded in dialysis buttons were then added to high salt dialysis buffer [10 mM Tris-HCl pH 7.5 at 4 °C, 1 mM EDTA, 2 M NaCl, 0.01 % Triton X-100, 5 mM β-mercaptoethanol] at 4 °C. The high salt dialysis buffer was then exchanged into low salt dialysis buffer [10 mM Tris-HCl pH 7.5 at 4°C, 1 mM EDTA, 50 mM NaCl, 0.01 % Triton X-100, 5 mM β-mercaptoethanol] at a rate of ~0.7 ml/min over a period of 30-40 h. Nucleosome quality was assessed on a 5 % 37.5:1 acrylamide/bis 0.5 x TBE gel (120 V).

Mass Photometry Analysis of Nucleosomes

Mass photometry analysis was performed on a Refeyn Two^{MP} instrument and analyzed using DiscoverMP v2022R1 (with IScat v1.52.0 and IScat Utils v1.43.0). 24 x 50 mm with 170 ± 5 µm thickness No. 1.5H glass slides (ThorLabs, Cat# CG15KH1) and silicone gaskets (Grace Bio-labs, Cat# 103250) were washed 2x with alternating 100 % isopropanol and deionized water. Clean gaskets were then placed on the cleaned slides to serve as a sample chamber. 5 µl of 1x PBS was placed in the gasket to calibrate each sample collection then 5 µl of nucleosome samples were added for a target final concentration of ~5-10 ng/µl nucleosomes in the well. Measurements were conducted at room temperature.

Cryo-EM Sample Preparation and Data Collection

Human H2A.Z nucleosomes were reconstituted on either 601L or Sat2R-P DNA as described above with a final concentration aim of 1.6 µM in 1 ml. Reconstituted nucleosomes were dialyzed into buffer containing 10 mM HEPES-KOH pH 7.4 and 30 mM KCl then concentrated to a final concentration of ~1 mg/ml using Amicon® Ultra Centrifugal Filters (Cat# UFC5010).

Samples were then mixed with 10x cryo-protectant buffer [10 mM HEPES-KOH pH 7.5, 30 mM KCl, 10 % Trehalose, 1 % Hexanediol, 1 mM MgCl₂] and 3 µl of each solution was applied to plasma-cleaned Quantifoil gold R1.2/1.3 400-mesh grids (Quantifoil) either graphene coated (graphene coating was done in-house as specified in (Arimura et al., 2025 [DOI](#)) or not for the Sat2R-P and 601L samples, respectively. Grids were then vitrified using the Vitrobot Mark IV (FEI) at 100 % humidity, 2 sec blotting time, blot force 1. Data collection was conducted using the Titan Krios (ThermoFisher), equipped with a 300 kV field emission gun and a K3 direct electron detector (Gatan), at a magnification of x81,000 (0.86 Å/pixel). Total micrographs collected for each sample as follows: 4544 for unmethylated Sat2R-P nucleosomes; 6891 for methylated Sat2R-P; 7020 for unmethylated 601L; 5427 for methylated 601L. Data statistics summarized in [Suppl Table 1](#) [DOI](#).

Cryo-EM Data Analysis

Diagrams of analysis workflows are shown in [Suppl Fig 1](#) [DOI](#) and [Suppl Fig 5](#) [DOI](#). Micrographs were motion-corrected using MotionCor2 (Zheng et al., 2017 [DOI](#)) in Relion v4 (Scheres, 2012 [DOI](#)). Downstream analysis was conducted in CryoSPARC v4.6 (Structura Biotechnology Inc.). Motion-corrected micrographs then underwent patch CTF correction. Particles were picked using Topaz v0.2 (Bepler et al., 2019 [DOI](#)) trained on a set of 1,000-2,000 manually picked nucleosome-like particles and then extracted to 384 pix (~330 Å) boxes with 4x binning and used for 2D classification into 200-250 classes. Particles from selected classes were then run through a series of heterogeneous refinements with 1-2 nucleosome-like models and 4 decoy models generated

through *ab initio* reconstruction to sort out junk particles from those containing nucleosomes. The heterogeneous refinements were iterated until greater than 90 % of particles in the round were classified as nucleosome-like. These particles were then further refined either via Refine 3D, CTF refinement, and polishing in Relion v4 (601L structures) or homogeneous and non-uniform refinements in CryoSPARC v4.6 (Sat2R-P). Particles for the methylated Sat2R-P structure underwent another heterogeneous refinement using a closed nucleosome, open, and hexasome model as starting classes to select for the highest resolved closed nucleosome structure. Final resolutions were determined using the gold standard FSC threshold (FSC = 0.143). Local resolution maps shown in Suppl Fig 3 were generated using the local resolution estimation tool with default parameters in CryoSPARC. The maps were then visualized on density maps filtered using the local filter tool with default parameters. For the *in silico* mixing 3D classification analysis (Suppl Fig 4): micrographs from both unmethylated and methylated Sat2R-P structures were merged and subjected to the data analysis pipeline described above. Six starting models representing different nucleosome structure types were then constructed using the *ab initio* 3D reconstruction tool. These models were then used as input volumes, and cleaned particles from the merged analysis were then sorted to each of these models using the heterorefinement tool in CryoSPARC.

Atomic Modeling and RMSD analysis

Initial atomic models were generated using previously published structures of human H2A.Z-containing histone core (PDB ID: 3WA9) (Horikoshi et al., 2013) and modified versions of either 601L DNA (PDB ID: 3UT9) (Chua et al., 2012b) or Sat2R DNA (PDB ID: 5CPI) (Osakabe et al., 2015). Models were docked into the EM density maps using Phenix v1.21, then iteratively refined and assessed in Phenix v1.21 (Liebschner et al., 2019) and Coot 0.9.8.7 (Emsley et al., 2010). RMSD values were calculated in PyMOL v2.5.7 (Schrödinger) using the cealign function. For the Sat2R-P models, DNA positioning was determined using the EM densities at the dyad of the unmethylated map which allowed for pyrimidine and purine distinction. A two-base pair shift was determined by assessing ambiguity of densities around nucleotides which change from purine to pyrimidine, and *vice versa*, between mirrored DNA models versus those which retain identity. After determining best-fit DNA models for the unmethylated structure, the same DNA positionings were then used for the methylated structure.

DNA Preparation for Restriction Digest Assay

Non-palindromic 1HinfI_Sat2R DNA (Table 1) was prepared via PCR using Q5 polymerase (NEB Cat# M0491). Final concentrations of reaction mixture as follows: 1x Q5 reaction buffer; 200 μ M dNTPs; 2 μ M forward and reverse primers; 0.2 g/ μ l DNA template; 0.02 U/ μ l Q5 polymerase. Roughly 60-65 PCR cycles were run following manufacturer's protocol with a melting temperature of 60 °C. PCR products were phenol-chloroform extracted and then FPLC purified in 1xTE buffer (Superdex 200 10/300 GL increase column, Cytiva, Cat# 28990944).

Restriction Digest Assay

Nucleosomes were mixed at 30 ng per timepoint sample in 1x rCutSmart® buffer (NEB Cat# B6004S) [50 mM potassium acetate, 20 mM Tris-acetate pH 7.9 at RT, 10 mM magnesium acetate, 100 μ g/ml recombinant albumin] and 0.5 U/ μ l HinfI restriction enzyme (NEB Cat# R0155S). Samples were incubated at 37 °C and added to stop buffer [final concentrations: 20 mM Tris-HCl pH 8.0 at RT, 20 mM EDTA, 0.1 % SDS, 0.6 mg/ml Proteinase K] at each timepoint. Stopped samples were then incubated at 55 °C for 40 min and run on a 10 % 37.5:1 acrylamide/bis 0.5x TBE gel (55 min at 150 V) and stained with SYBR™ Safe DNA stain (Thermo Fisher Scientific, Cat# S33102). Fluorescent signals were measured on a LI-COR Odyssey M system and quantified using ImageJ via histogram analysis as previously described in (Stael et al., 2022). The ratio of Cy5: SYBR™ Safe signals was then determined for each time point and plotted. A two-tailed t-test assuming unequal variance was then performed on data from n = 5 experiments for statistical analysis.

Antibodies

Antibodies used in this study are listed in Table 2 [↗](#). Custom antibodies were generated through Cocalico Biologicals, Inc.™ using their standard protocol. Target binding of the H2A.Z antibody used for CnT-BS and SRCAP custom antibody in *Xenopus* egg extract was verified using mass spectrometry analysis (MS) (Supplementary Data 1 and 2 [↗](#)).

Mass Spectrometry Validation

For the MS analysis, 20 µl of antibody-bound Protein A Dynabeads™ (Invitrogen, Cat# 10008D) (0.8 µg antibody per 1 µl of beads) were incubated at a 1:1 ratio with CSF egg extract (see next section) at 4 °C for 1h. Samples were collected via magnet and washed 3 times with sperm dilution buffer (SDB) [150 mM Sucrose, 10 mM HEPES-KOH pH 8.0, 1 mM MgCl₂, 100 mM KCl] then run on a 4-20% gradient SDS-PAGE gel as a 1 cm short-stack. The gel was then stained with Coomassie Brilliant Blue G-250 (Thermo Fisher) and the short stack was cut out. The cut gel was then destained using a solution of 30 % acetonitrile and 100 mM ammonium bicarbonate in water and dehydrated using 100 % acetonitrile. Samples were subjected to heat assisted (50 °C) trypsinization for 1 h. The resulting peptides were extracted three times with a solution of 70 % acetonitrile and 0.1 % formic acid then concentrated using a SpeedVac and resuspended in 10 µl volumes. 3 µl of purified peptides were subsequently analyzed by LC-MS/MS using a Dionex 3000 HPLC system equipped with an EasySprayer (ES902), coupled to an Orbitrap ASCEND mass spectrometer from Thermo Scientific. Peptides were separated by reversed-phase chromatography using solvent A (0.1 % formic acid in water) and solvent B (80 % acetonitrile, 0.1 % formic acid in water) across a 45-min gradient. Data was processed using ProteomeDiscoverer 1.4 and quantified using ProteomeDiscoverer v1.4/Mascot v3.1. The spectra were then queried against a *Xenopus laevis* database (Wühr et al., 2014a [↗](#)) and matched peptides filtered using a Percolator calculated FDR of 1 %. As proxy for protein amount, the average area of the three most abundant peptide signals was used. Relative enrichment of proteins in the sample was then calculated by dividing by known protein concentrations in *Xenopus laevis* eggs and then the enrichment over IgG control was calculated. Common contaminants were marked in red.

Xenopus laevis frogs and preparation of sperm and CSF egg extract

Female *Xenopus laevis* frogs were purchased from Xenopus 1. Male frogs were purchased from either Xenopus 1 (for use in non-sequencing experiments) or the National Xenopus Resource (*XLae-J-Strain*^{NXR}, Cat# NXR_0024; for use in sequencing experiments). Vertebrate animal protocols (20031 and 23020) approved by the Rockefeller University Institutional Animal Care and Use Committee were followed for housing and care.

CSF egg extract and demembranated sperm nuclei were prepared as specified previously (Tutter and Walter, 2006 [↗](#)).

Sperm Pronuclei Preparation

Sperm from J-strain frogs were added to prepared CSF egg extract at a final concentration of 1,500 sperm/µl extract with a total extract volume of 200 µl for each sample. 90 ng (~1 % of total DNA) of lambda DNA (Promega Cat# D1521) was also added to assess bisulfite conversion efficiency and the extract solution was then cycled to interphase with the addition of CaCl₂ to a final concentration of 0.3 mM and incubated at 20 °C for 90 min. Samples were then diluted 1:4 with 1xDB3 buffer [10 mM HEPES-KOH pH 8.0, 100mM NaCl, 0.5 mM spermine, 0.5 mM spermidine, 1 mM ASEBF (GoldBio Cat#A-540-1), 1x ROCHE cOmplete™ protease inhibitor cocktail (Cat# 11697498001), 200 mM Sucrose, 0.1 % Tween-20], layered onto 1 ml of 60SDB3 [1xDB3 + 60 % sucrose] and centrifuged at 4 °C, 10,000 rpm for 45 min (Beckman Coulter SX241.5 rotor). Sample pellets were washed 2x with 1xDB3, centrifuging at 4,000 x g, 4 °C for 4 min, then resuspended in 1x CnT-WB [20 mM HEPES-KOH pH 7.8, 150 mM NaCl, 0.5 mM spermidine, 1x ROCHE cOmplete™ protease inhibitor cocktail, 0.1 % Tween-20].

Antigen	Host	Company	Catalog No.	Peptide Used for Generation	Dilution	Assay(s) Used In
H2A.Z	Rabbit	Custom		CIHKSLIGK KGQQKTV	2 µg/ml	Western Blots
H2A.Z	Rabbit	Active Motif	Cat#39013		1:10	CnT-BS
H3	Mouse	Cosmo Bio USA	MABI0001-100-EX		1:100	CnT-BS
H4	Mouse	Active Motif	Cat# 61521		1:1000	Western Blots
H2A.X-F	Rabbit	Custom, gift from David Shechter		Shechter et al. 2009	1:1000 (of serum)	Western Blots
Ku70	Rabbit	Custom		Postow et al. 2008	2 µg/ml	Western Blots
SRCAP	Rabbit	Custom		CPNSPSSH RVRKAKT	0.8 µg per 1 µl of beads (depletions); 12 µg/ml (WB)	Depletions; Western Blots
ZNHIT1	Rabbit	Proteintech	16595-1-AP		1:1000	Western Blots
P400	Rabbit	Bethyl	A300-541A		1:1000	Western Blots
ALFA tag	Alpaca (single domain antibody purified from E. coli)	NanoTag Biotechnologies	N1502-Li800-L		1:1000	Western Blots
<i>Secondaries</i>						
IRDye® 800CW anti-Rabbit IgG	Goat	LICORbio	926-32211		1:10000	Western Blots
IRDye® 680RD anti-Mouse IgG	Goat	LICORbio	926-68070		1:10000	Western Blots
anti-Rabbit IgG	Guinea Pig	antibodies-online.com	ABIN101961		1:100	CnT-BS
anti-Mouse IgG	Rabbit	abcam	ab46540		1:100	CnT-BS

Table 2. Antibodies used in this study.

XTC-2 Cell Line

XTC-2 cells (gift from Dr. David Shechter) were cultured in Leibovitz's L-15 Medium (Gibco™, Cat# 11415064) (0.7x amphibian strength, diluted with cell culture grade water (Corning®, Cat# 46-000-CV)) supplemented with 500 U/ml each of penicillin and streptomycin (Gibco™, Cat# 15140122), and 10 % FBS (R&D systems, Cat# S11150). XTC-2 cells were cultured in a 26 °C incubator without CO₂ supply. Cell line identity was verified via mitotic-spread karyotyping.

XTC-2 Nuclei Preparation

XTC-2 cells were harvested at 90-100 % confluency from a 15 cm dish. Nuclei isolation was performed at 4 °C using the Nuclei EZ Prep kit (Millipore Cat# NUC101-1KT) as per manufacturer's protocol and resuspended in 1x CnT-WB. For each sample, 18 ng (~1 % of total DNA) of unmethylated lambda DNA (Promega Cat# D1521) was added to 300,000 nuclei to assess for bisulfite conversion efficiency.

Purification of Tn5

Tn5 was purified as described in (Soroczynski et al., 2024 [\[link\]](#)). Briefly, VR124 pETv2-10His-pG-Tn5 (E54K, L372P) *E. coli* strain (Addgene #198468) were cultured to log growth phase and Tn5 expression induced with IPTG addition. Cells were harvested at 8,000 x g, 30 min at 4 °C, washed with ice-cold DPBS, re-pelleted and frozen. Frozen pellets were then resuspended at 4 °C in lysis buffer LysEQ [20 mM HEPES pH 7.8, 800 mM NaCl, 20 mM imidazole, 10 % glycerol, 1 mM EDTA, 2 mM TCEP-HCl, 1x EDTA-free cOmplete protease inhibitor] and sonicated. Lysates were pelleted at 20,000 x g, 35 min at 4 °C, then filtered through a 0.45 µm PES filter. Supernatant was then loaded onto a HisTrap HP 5 ml column (Cytiva, Cat# 17-5247-01) pre-equilibrated with LysEQ buffer. The loaded column was then washed with 10 column volumes each of WashB1 [20 mM HEPES pH 7.8, 800 mM NaCl, 30 mM imidazole, 10 % glycerol, 1 mM EDTA, 2 mM TCEP-HCl] and WashB2 [20 mM HEPES pH 7.8, 800 mM NaCl, 45 mM Imidazole, 10 % glycerol, 1 mM EDTA, 2 mM TCEP-HCl]. Protein was then eluted with EluB [20 mM HEPES pH 7.8, 800 mM NaCl, 400 mM imidazole, 10 % glycerol, 1 mM EDTA, 2 mM TCEP-HCl]. Tn5-containing fractions were pooled and dialyzed 4 °C overnight in SECB buffer [50 mM Tris-HCl pH 7.5, 800 mM NaCl, 10 % glycerol, 0.2 mM EDTA, 2 mM DTT] then filtered through a 0.45 µm PES filter and FPLC purified on a HiLoad 16/600 Superdex 200 pg column (Cytiva, Cat#: 28989335). Purified fractions were pooled and concentrated with a 30 MWCO Amicon Ultra Centrifugal filter (final concentration 12 µM) and mixed with UltraPure glycerol to a final concentration of 55 % glycerol. Purified Tn5 was stored at -20 °C.

CUT&Tag-Bisulfite Library Preparation

Sequencing libraries were prepared as previously described in (Li et al., 2021 [\[link\]](#)). Briefly, prepared nuclei samples were bound to 10 µl of activated concanavalin A beads (EpiCypher Cat# 21-1401) and incubated with specified primary and secondary antibodies (Table 2 [\[link\]](#)) in 50 µl Antibody Binding Buffer [CnT-WB + 2 mM EDTA + 0.1 % w/v BSA] for 1 h shaking at RT, washing 3x with CnT-WB between each incubation. Samples were then incubated with 50 µl of transposome complex (1:1 pG-Tn5:adaptor primer mix, ~0.1-0.2 µM final to each sample) in CnT-300 [CnT-WB with 300 mM NaCl] for 1 h at RT then washed with CnT-300. Transposome activity was initiated with addition of 300 µl Tagmentation Buffer [20 mM HEPES-KOH pH 7.8, 300 mM NaCl, 0.5 mM spermidine, 10 mM MgCl₂, 1x ROCHE cOmplete™ protease inhibitor cocktail, 0.1 % Tween-20] and incubated 37 °C, 1500 rpm on thermomixer, 1 h. Reactions were stopped with addition of stop buffer [15 mM EDTA, 0.1 % SDS, 0.2 mg/mL Proteinase K final concentrations] and incubated at 55 °C overnight, then phenol-chloroform extracted. Purified samples were resuspended in 11 µl of 10 mM Tris-HCl, pH 8.0 and oligonucleotide replacement/gap repaired carried out through addition of 2 µl of 10 µM Tn5mC-ReplO1 DNA oligo (Table 1 [\[link\]](#)), 2 µl 10x Ampligase Buffer (Lucigen), 2 µl dNTP mix (2.5 mM for each dNTP). Solutions were subject to the following heating protocol: 50 °C 1 min, 45 °C 10 min, cool to 37 °C at -0.1 °C/second. Once at 37 °C, 3U of T4 DNA polymerase and 12.5U of Ampligase (Lucigen) were added and samples incubated for 30 min. Reactions were stopped with 20 mM EDTA final and purified using the MinElute PCR purification kit (Qiagen Cat# 28004).

Samples were bisulfite converted using the EZ DNA Methylation-Lightning Kit (Zymo Research Cat# D5030) and libraries amplified (14 cycles) using NEBNext Q5U Master Mix (NEB Cat# M0597) and Illumina Nexera i5/i7 adapter primers.

Sequencing Data Analysis

Adapter trimming, deduplication, and genome alignment

Libraries were sent to Novogene for sequencing on the NovaSeqX Plus system (PE150). XTC-2 data was then downsampled to 30 million reads using seqtk (v1.5-r133) to have all samples at similar read depths. Adapter trimming and deduplication of the demultiplexed reads was then done using fastp v0.24.0 (Chen, 2023 [\[1\]](#)) [`--adapter_sequence CTGTCTCTTATACAC --adapter_sequence_r2 CTGTCTCTTATACAC -g --poly_g_min_len 5 -p -l 20 --cut_tail -y -Y 20 -q 20 --dedup`]. Processed reads were then aligned to the *Xenopus laevis* genome (Xenla 10.1) and methylated CpG proportions for all mapped reads calculated using Bismark v0.24.2 (Krueger and Andrews, 2011 [\[2\]](#)) [`-q --local --score-min G,20,8 --ignorequals --no-mixed --no-discordant --dovetail --maxins 700`].

Count matrix generation and sequencing depth normalization

The *Xenopus laevis* genome was subset into 1000 bp bins. Reads from processed bam files were then tallied according to bins to generate a genome-wide count matrix. Mitochondrial associated reads and bins which contained no reads across all samples were removed. Size factors were then estimated using DESeq2 v1.42.1 (Love et al., 2014 [\[3\]](#)) to generate sequence depth normalized bigwig files. Count matrices were also vst transformed in DESeq2 then used to generate the read fragment correlation plot. For other analyses, only genomic bins containing at least 17 reads across at least 2 samples were kept and replicates for each condition were averaged.

DNA Methylation analysis

To get methylated CpG proportions for each sample (Fig 4A [\[4\]](#)), H2A.Z peaks were first identified with SEACR (v1.3) (Meers et al., 2019 [\[5\]](#)) using a cutoff of 0.025 by AUC with a stringent threshold. Individual CpG data was then extracted using Bismark with a cutoff of 5 reads and disregarding the first 9 bp of read 2 (due to adapter fill-in). Methylation proportions for H2A.Z samples were then calculated for CpGs overlapping with H2A.Z peaks. For H3, only regions outside of identified H2A.Z peaks were used to assess the CpG methylation status of general genomic background non-coincident with H2A.Z enriched areas. To generate H3-associated methylation 500 bp tiles for use in plotting (Suppl Fig 9B-D [\[6\]](#)), methylation data was extracted with a specified cutoff of 1 read and disregarding the first 9 bp of read 2 then imported into a methylRawList object with a minimum read coverage of 2 and counted over 500 bp tiles with a 500 bp step size and a minimum of 2 CpG sites using methylKit v1.28.0 (Akalin et al., 2012 [\[7\]](#)).

Generation of plots

Genomic annotations for the top 20 % of filtered bins containing the most reads for each condition were generated and proportions plotted using ChIPseeker v1.38.0 (Yu et al., 2015 [\[8\]](#)) with reference gene annotations gathered from Xenbase (Fisher et al., 2023 [\[9\]](#)). Heat maps and plot profiles were created using deepTools v3.5.5 (Ramírez et al., 2016 [\[10\]](#)). For comparison of H3 methylation sites and H2A.Z, a signal matrix was first generated using the tiled H3 methylation bigwigs against all annotated *Xenopus laevis* genes, removing genes with no methylation data available. The resulting gene list was then used to plot corresponding H2A.Z signals.

in vitro mRNA Synthesis

DNA sequences of proteins of interest were cloned into pSP64T plasmids modified to include a Kozak sequence (GCCACC) upstream of the coding sequence and linearized. pSP64T (DM#111) was a gift from Douglas Melton (Addgene Plasmid# 15030). RNA production was carried out using the MEGAscript™ SP6 transcription kit (Invitrogen, Cat# AM1330) as per manufacturer's protocol, with the addition of 4.5 mM CleanCap® Reagent AG (TriLink, Cat# N-7113) and a reduction in GTP concentration down to 1.5 mM final.

Chromatinization in Egg Extract Assays

pAS696 containing a 19-mer array of Widom 601 DNA was a gift from Aaron Straight (Guse et al., 2011). 601 and HSat2 arrays were digested with XbaI and EcoRI or XbaI and BamHI, respectively, along with DraI and HaeII. DNA fragments containing the array sequence were then purified through a PEG precipitation titration, starting at 4.5 % of PEG (MW8000) and increasing in 0.5 % increments until 8 % (NaCl added to 0.5 M final concentration in the starting solution). Fragment containing fractions were then assessed via agarose gel and purified via phenol-chloroform extraction. Fragment ends were biotinylated with biotin-14-dATP (Jena Bioscience, Cat# NU-835-BIO14-S) using Klenow Fragment (3' → 5' exo-) (NEB Cat# M0212) following manufacturer's protocol. Biotinylated DNA was phenol-chloroform extracted and bound to M-280 Streptavidin Dynabeads™ (Invitrogen, Cat# 11206D) at 300 ng DNA per 1 µl of beads. DNA bead binding was carried out at room temperature for a minimum of 1 h shaking in the following solution: 1x DNA Bead Binding buffer (DBB) [750 mM NaCl, 50 mM Tris-HCl pH 8.0, 0.25 mM EDTA, 0.05 % Triton-X-100]; 2.5 % (w/v) polyvinyl alcohol; 750 mM NaCl. DNA beads were then washed 2x with DBB and 3x with sperm dilution buffer (SDB) [150 mM Sucrose, 10 mM HEPES-KOH pH 8.0, 1 mM MgCl₂, 100 mM KCl] and added to egg extract at a ratio of 4 µl DNA beads to 30 µl of egg extract. Experiments that did not use mRNA for exogenous protein expression also included 0.1 mg/ml cycloheximide dissolved in EtOH. CaCl₂ was then added to the extracts at a final concentration of 0.3 mM and incubated at 20 °C for 1 h. Samples were then diluted 10-fold with SDB, and DNA beads were collected on a magnetic rack and washed 3x with SDB + 0.05 % Triton-X-100 before running on SDS-PAGE gel and assessed via western blot.

Western Blot Analysis

Samples were boiled in 1x SDS loading buffer [125 mM Tris-HCl pH 6.8, 0.01 % SDS, 25 % glycerol, 0.03 mg/ml bromo-phenol blue, 350 mM 2-mercaptoethanol] then run on either a 4-20 % SDS-PAGE gel (Bio-Rad, Cat# 4561096) or 3-8 % Tris-acetate gel (for SRCAP detection) (Invitrogen, Cat# EA0375BOX). Samples were then transferred from the gel to a nitrocellulose membrane (Cytiva, 0.1 µm pore size, Cat# 10600000) overnight at 4 °C in 1x transfer buffer [20 mM Tris, 150 mM Glycine, 20 % Methanol] (14V). Membranes were blocked for 30 min at RT with Intercept® (TBS) Protein-Free Blocking Buffer (LICORbio, Cat# 927-80001), followed by primary antibody incubation (diluted in Intercept® T20 (TBS) Antibody diluent (LICORbio, Cat# 927-65001)) for 1 h at RT. Membranes were then washed 3x, 5 min each wash, in 1xTBS+0.05 % Tween-20 then incubated with appropriate secondary antibodies for 1 h at RT and washed. Blots were then imaged on a LI-COR Odyssey M system and signals quantified using ImageJ via histogram analysis as previously described in (Stael et al., 2022). Ratios of either H2A.Z or H2A.X-F signals to H4 signals were then determined for normalization purposes in the chromatinization assays.

SRCAP Depletion from Egg Extract

SRCAP antibody was bound to Protein A Dynabeads™ (Invitrogen, Cat# 10008D) at a concentration of 0.8 µg antibody per 1 µl of beads overnight at 4 °C in wash/coupling (W/C) buffer [10 mM HEPES-KOH pH 8.0, 150 mM NaCl]. Beads were then washed 2x with W/C buffer and 3x with SDB. An equal volume of beads and egg extract (supplemented with 0.1 mg/ml cycloheximide) were then incubated together for 1 h at 4 °C and beads removed via magnet.

Other Instruments and Software Used

ÄKTA pure SN2626712 (Cytiva) and UNICORN v7.11 (Cytiva) for FPLC purification of histone proteins.

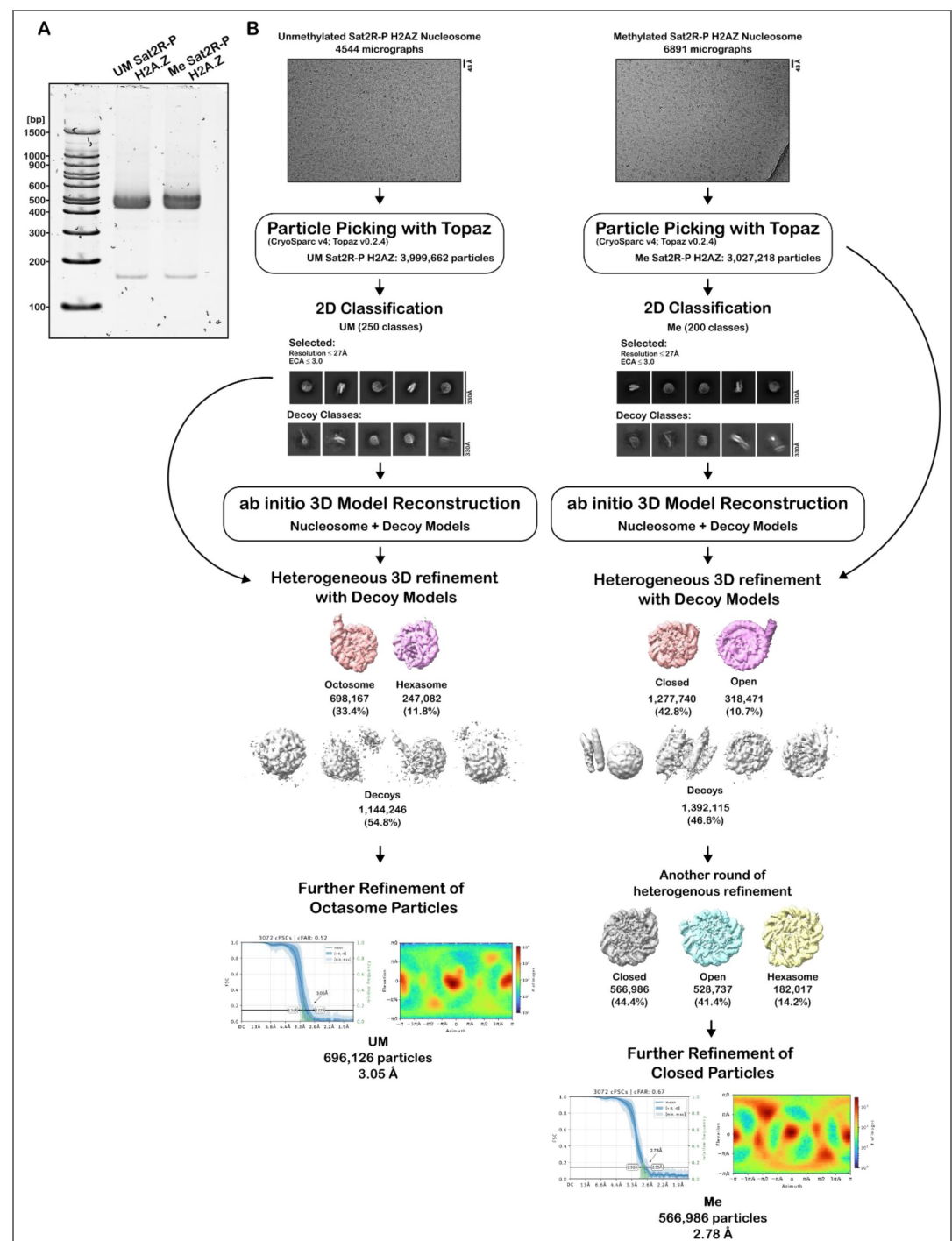
UCSF ChimeraX v1.9 (Meng et al., 2023) for structure visualization.

LI-COR Odyssey M, LI-COR Acquisition v1.2, and ImageJ v1.54f (Schneider et al., 2012) for western blot and gel imaging, data analysis, and visualization.

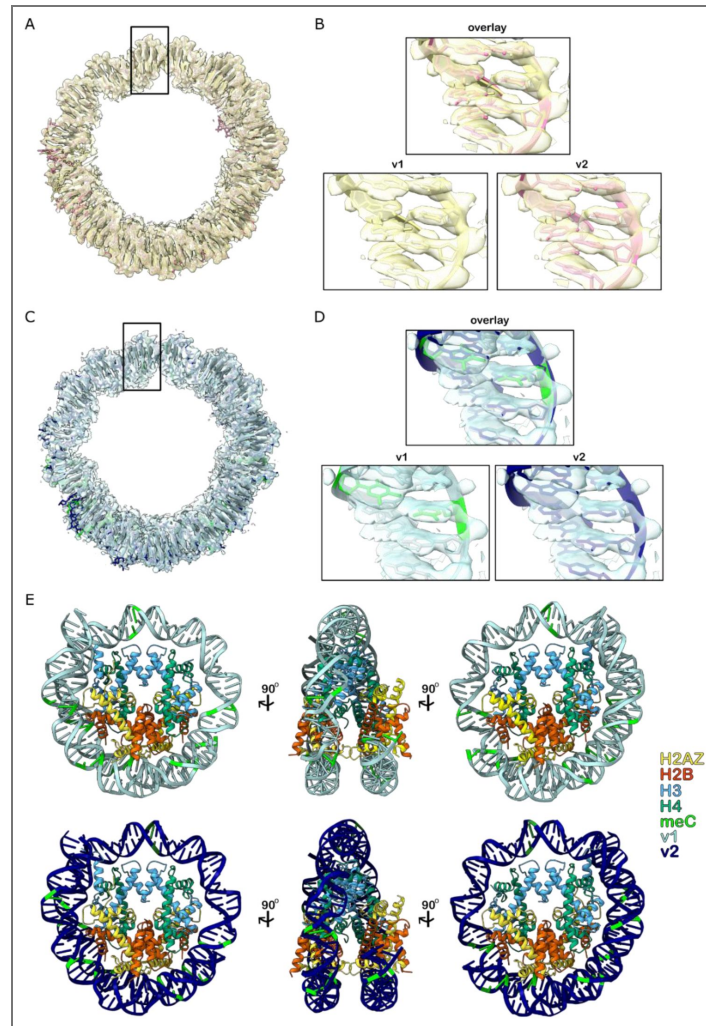
QuantStudio™ 5 (Thermo Fisher, Cat# A28574) and QuantStudio™ Design & Analysis v2.7 for QC of sequencing samples.

Inkscape v1.3.2 for figure generation.

Supplementary figures and tables



Supplementary Figure 1. Workflow of Sat2R-P cryo-EM analysis. (A) Native PAGE analysis of H2A.Z nucleosomes used for cryo-EM analysis. Bands visualized through SYBR Safe staining. (B) Diagram of structure analysis pipeline for either unmethylated (left) or methylated (right) H2A.Z SAT2R-P samples. Further details described in methods.

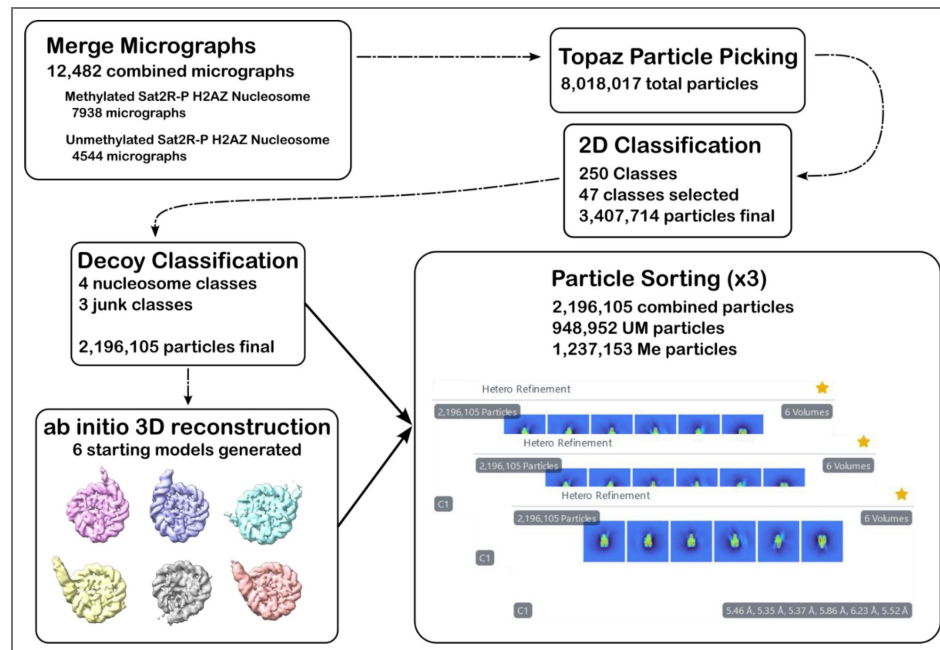


Supplementary Figure 2. DNA atomic model generation for Sat2R-P H2A.Z nucleosome structures.

(A and C) Overlays of electron density maps for the DNA with both generated atomic models for either the unmethylated **(A)** or methylated **(C)** Sat2R-P H2A.Z nucleosome structure. **(B and D)** Zoom-in of dyad regions showing fitting of both DNA atomic models (v1 and v2) generated for either the unmethylated **(B)** or methylated **(D)** structure. Methylated cytosines are highlighted in green. Top panels are overlays of both models for the two structures. Positioning was determined by assessing densities around bases which retain purine/pyrimidine identity in both models vs those that switch. Bases that switched from purine to pyrimidine, and vice versa, showed more ambiguous densities than those that retained identity. Positioning was determined using the unmethylated structure and then used in modeling the methylated structure. **(E)** Comparison of predicted CpG positioning for the v1 (top)

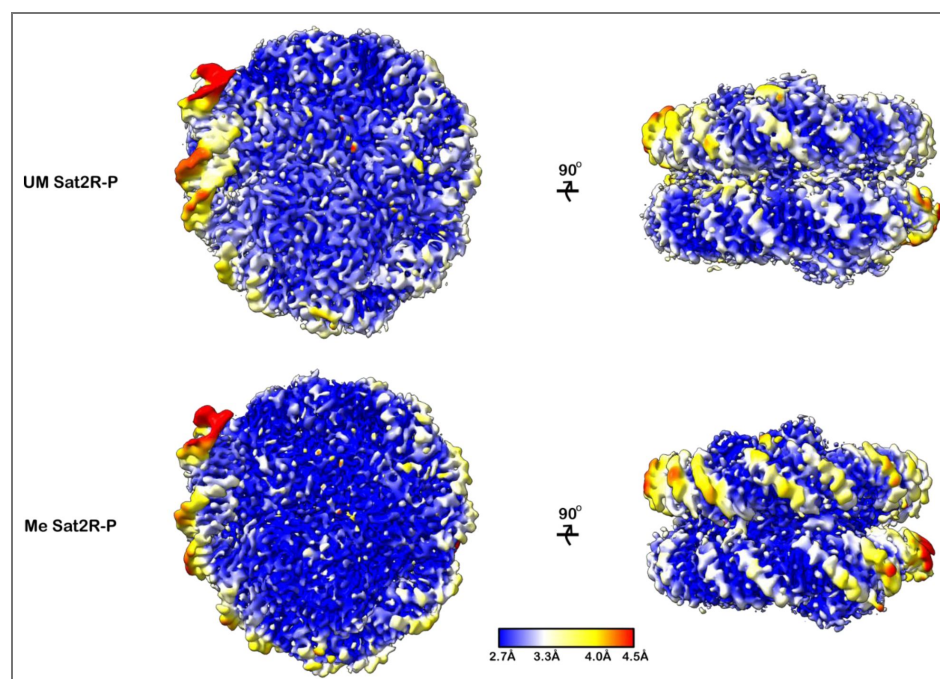
Supplementary Figure 3. Workflow of *in silico* mixing 3D classification analysis.

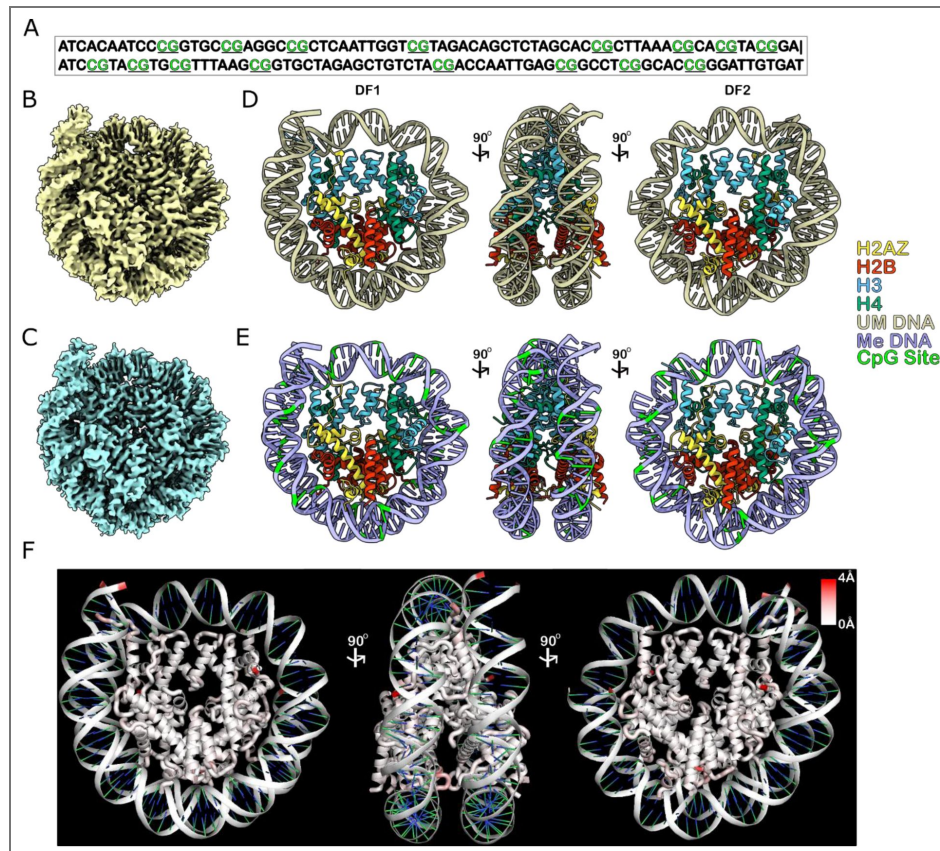
Particles from a merged batch of unmethylated and methylated Sat2R-P H2A.Z nucleosome micrographs were picked and fed through the standard analysis pipeline. Six 3D models were then generated representing various open, closed, or hexosome-containing states of the nucleosome. Merged particles were then sorted to each of the classes through CryoSPARC's heterorefinement tool (n = 3 technical replicates).



Supplementary Figure 4. Local resolution estimates for unmethylated and methylated Sat2R-P H2A.Z structures.

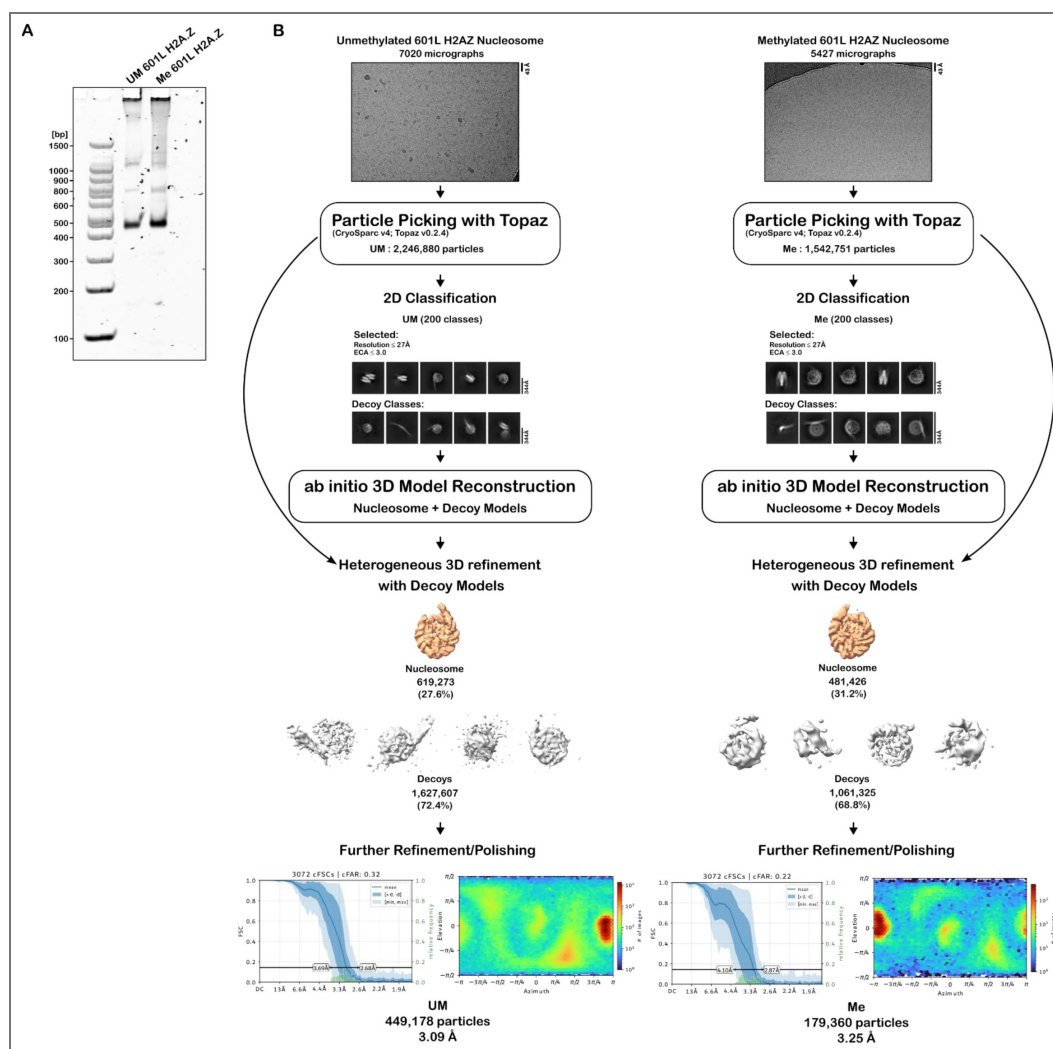
Left, local resolution estimates of DF1 for both unmethylated (top) and methylated (bottom) Sat2R-P density maps. Right, side view showing local resolution of the DNA backbone for both methylated and unmethylated density maps. Local resolution maps were generated in CryoSPARC using the local resolution estimation tool and overlaid on density maps filtered using CryoSPARC's local filtering tool.





Supplementary Figure 5. Cryo-EM structures of 601L H2A.Z nucleosomes show no DNA methylation dependent differences.

(A) Diagram of the palindromic 601L sequence. Bar indicates halfway point. Methylated cytosines are in green and underlined. (B and C) Final density maps of unmethylated (B) or methylated (C) H2A.Z nucleosomes. (D and E) Atomic models generated for (B and C). Methylated cytosines shown in green (F) RMSD analysis comparing differences between methylated and unmethylated 601L H2A.Z atomic models.

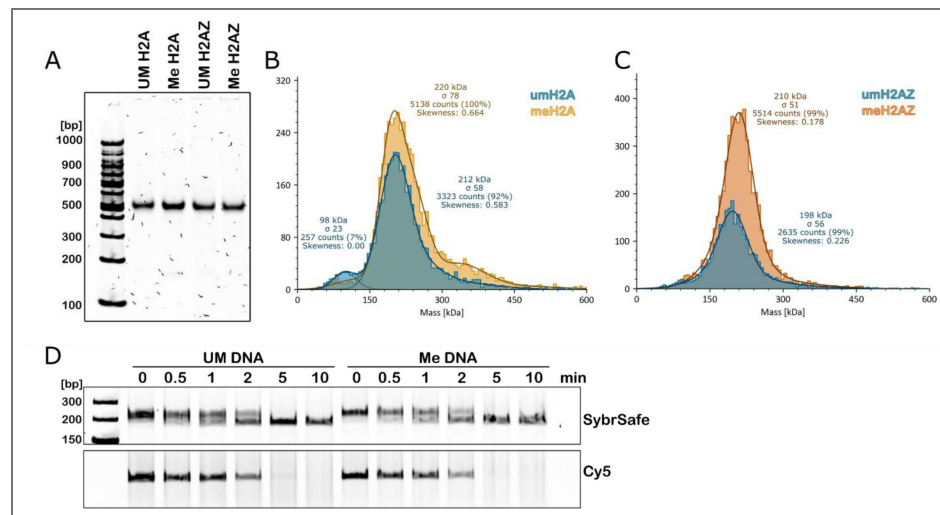


Supplementary Figure 6. Workflow of 601L H2A.Z nucleosome cryo-EM structures.

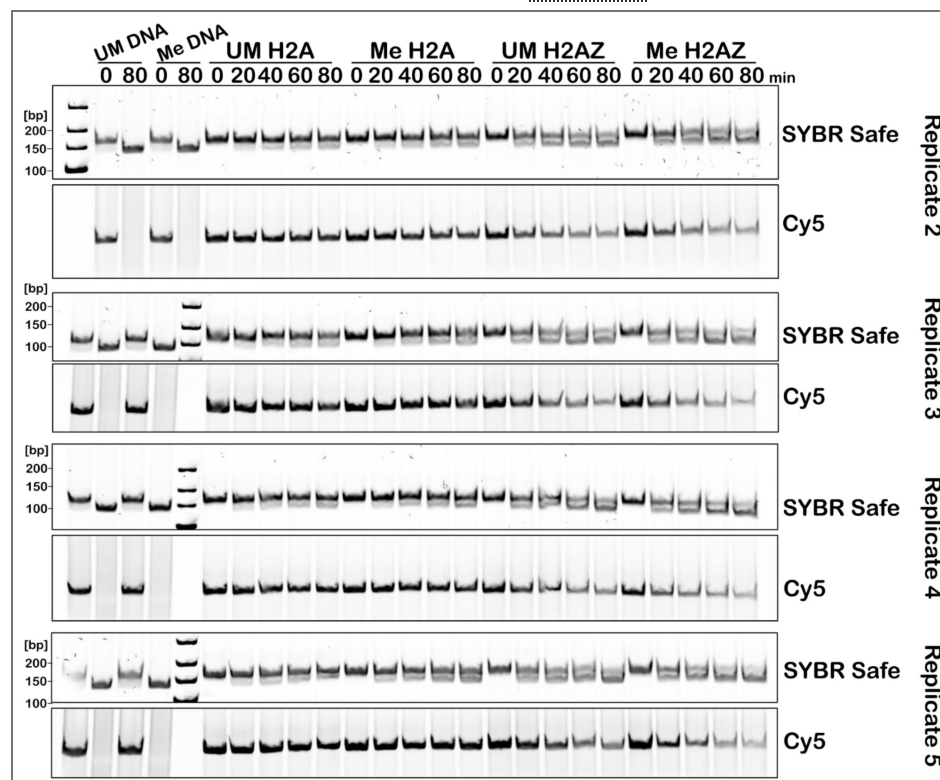
(A) Native PAGE analysis of 601L H2A.Z nucleosome samples used for cryo-EM. Bands visualized through SYBR Safe staining.
(B) Diagram showing analysis pipeline for solving 601L nucleosome structures. Further details described in methods.

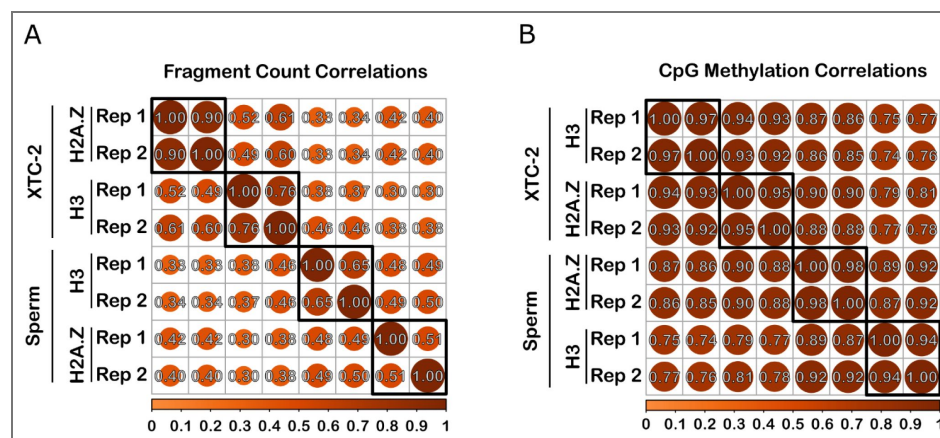
Supplementary Figure 7. Native PAGE and mass photometry analysis of nucleosomes used in *HinfI* digest assays.

(A) Native PAGE analysis of nucleosome samples. Bands visualized through SYBR Safe staining. (B and C) Mass photometry analysis of methylated or unmethylated H2A (B) or H2A.Z (C) nucleosomes. Peaks around 200-220 kDa represent the fully formed nucleosome population. (D) *HinfI* digest time course of bare unmethylated or methylated DNA.



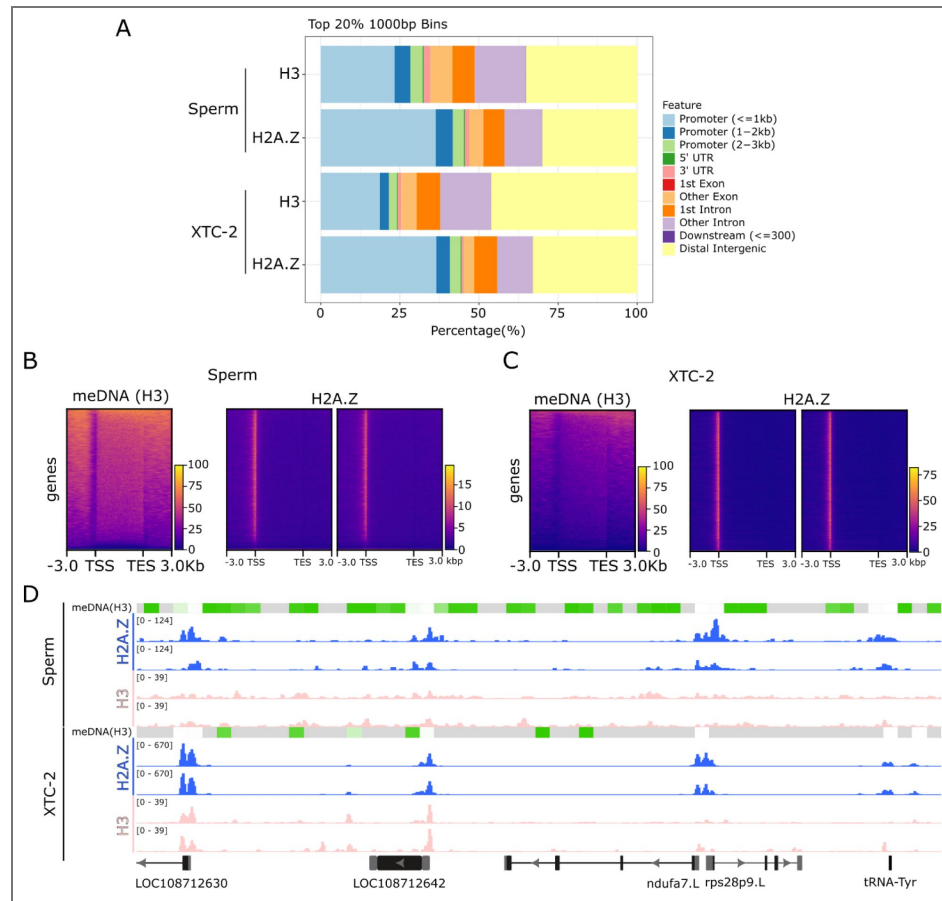
Supplementary Figure 8. Replicate gels of data from Main Figure 3.





Supplementary Figure 9. Reproducibility comparison of CnT-BS samples

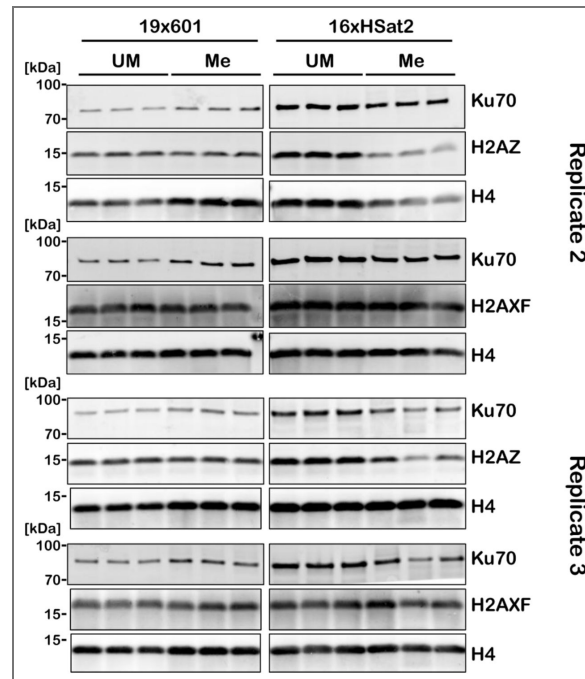
(A) Pearson correlation plot comparing fragment counts mapped to 1000 bp genomic bins across samples. Bins containing zero fragments were first filtered out. Fragment counts were then normalized using DESeq2 to account for sequencing depth differences and underwent a variance stabilizing transformation (VST) before the correlation matrix was generated. **(C)** Pearson correlation plot comparing CpG methylation across samples. Percent methylation at individual CpG sites was calculated for high coverage CpGs (≥ 10 reads) and normalized using methylKit. The correlation matrix was then generated using pairwise observations across all eight samples. Both correlation matrices were ordered, and rectangles drawn ($n = 4$, user-specified), via hierarchical clustering.



Supplementary Figure 10. H2A.Z localizes to hypomethylated TSSs.

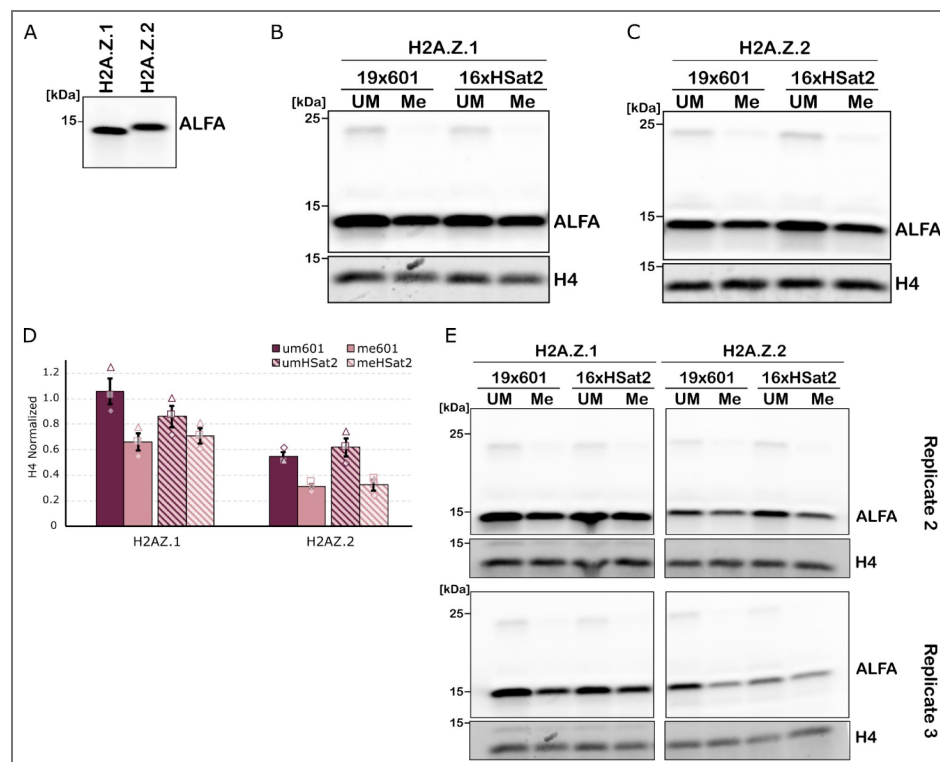
(A) Genomic annotation plots for the top quantile of 1000 bp genomic bins containing the most mapped reads averaged between replicates for each specified condition. **(B and C)** Heatmaps of either H3 CpG methylation calls averaged over 500 bp tiles (left-most panel) or H2A.Z reads (right panels) from two biological replicates across a representation of all annotated *Xenopus laevis* genes from either sperm **(B)** or XTC-2 **(C)** samples. Corresponding H3 and H2A.Z data are shown on the same sets of genes sorted in the same order. **(D)** IGV snapshot of a region on chromosome 1L showcasing preferential H2A.Z deposition at hypomethylated TSSs. Average H3 methylation tracks were generated across 500 bp tiles for visualization purposes on a scale of 0 to 100% CpG methylation. Grey indicates regions with no methylation data availability. Two biological replicates of H2A.Z and H3 tracks are shown.

Supplementary Figure 11. Replicates of data from Main Figure 4



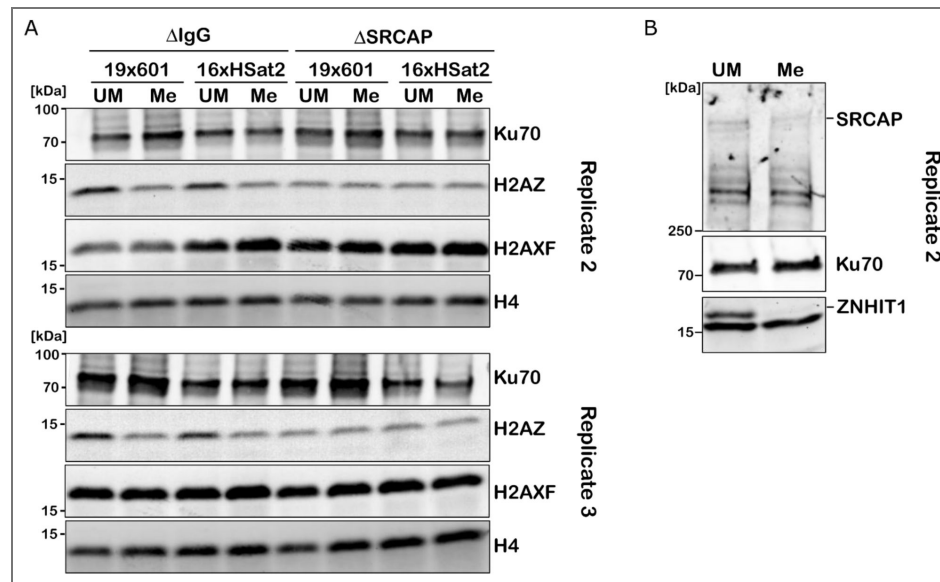
Supplementary Figure 12. Both H2A.Z variants preferentially associate with unmethylated DNA substrates.

(A) Expression check of exogenously added mRNA expression for ALFA-tagged versions of both H2A.Z.1 and H2A.Z.2. (B and C) Blots of DNA beads incubated with interphase extract expressing either ALFA-tagged H2A.Z.1 (B) and H2A.Z.2 (C). Presence of the specified H2A.Z variant on DNA beads was determined by blotting against the ALFA-tag with H4 shown as a loading control. (D) Quantification of (B and C). ALFA signals were normalized to H4 signals. Unique shapes denote results from three independent experiments (n = 3 biological replicates). Error bars represent SEM. (E) Additional replicate blots of data from (D).



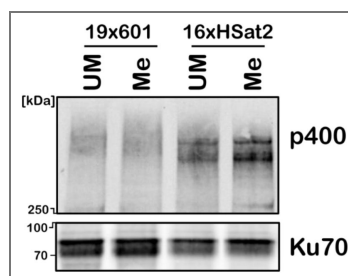
Supplementary Figure 13. Replicates of data from Main Figure 5

(A) Replicate blots of Main Figure 5B-D. (B) Replicate blots of Main Figure 5E-F with Ku70 as a loading control.



Supplementary Figure 14. Tip60-C does not display DNA methylation sensitivity.

Western blots of DNA bead pulldowns from interphase egg extract probing for endogenous p400 (ATPase for Tip60-C) along with respective loading controls.



Sample	UM Sat3R-P	Me Sat2R-P	UM 601L	Me 601L		
Data Collection						
Microscope	Titan Krios					
Magnification	81,000					
Voltage (kV)	300					
Camera	Gatan K3					
Electron exposure (e-/Å)	50.703		50			
Defocus range (µm)	-1.5 ~ -3.5					
Pixel size (Å)	0.86					
Micrographs Collected	4544	6891	7020	5427		
Data Processing						
Particles (no.)	696,126	566,986	449,178	179,360		
Symmetry imposed	C1	C1	C1	C1		
Map resolution (Å, FSC = 0.143)	3.05	2.78	3.09	3.25		
EMDR ID	EMD-70474	EMD-70473	EMD-70480	EMD-70481		
	UM Sat2R-P		Me Sat2R-P		601L	
Atomic Models	v1	v2	v1	v2	UM	Me
Composition						
Chains	10	10	10	10	10	10
Atoms	11117 (Hydrogens: 0)	10905 (Hydrogens: 0)	10585 (Hydrogens: 0)	10558 (Hydrogens: 0)	11101 (Hydrogens: 0)	11477 (Hydrogens: 0)
Residues	Protein: 735 Nucleotide: 259	Protein: 721 Nucleotide: 254	Protein: 701 Nucleotide: 246	Protein: 698 Nucleotide: 246	Protein: 722 Nucleotide: 264	Protein: 738 Nucleotide: 274
Water	0	0	0	0	0	0
Ligands	0	0	0	0	0	0
Bonds (RMSD)						
Length (Å) (# > 4σ)	0.004 (0)	0.003 (0)	0.004 (0)	0.003 (0)	0.003 (0)	0.004 (0)
Angles (°) (# > 4σ)	0.580 (0)	0.525 (3)	0.617 (0)	0.619 (0)	0.578 (1)	0.650 (0)
MolProbity score	1.78	1.47	1.77	1.82	1.6	2.35
Clash score	5.17	4.66	5.86	7.13	8.64	9.79
Ramachandran plot (%)						
Outliers	0	0	0	0	0.14	0
Allowed	2.5	1.56	1.9	2.35	1.56	4.16
Favored	97.5	98.44	98.1	97.65	98.3	95.84
Rama-Z (Ramachandran plot Z-score RMSD)						
whole (N = 719)	1.58 (0.32)	1.80 (0.32)	1.67 (0.32)	1.84 (0.32)	1.99 (0.32)	0.95 (0.30)
helix (N = 518)	2.25 (0.23)	2.33 (0.23)	2.07 (0.23)	2.20 (0.23)	2.25 (0.22)	1.63 (0.22)
sheet (N = 0)	--- (---)	--- (---)	--- (---)	--- (---)	--- (---)	--- (---)
loop (N = 201)	-2.34 (0.36)	-2.21 (0.36)	-2.05 (0.39)	-1.87 (0.38)	-1.43 (0.42)	-2.12 (0.36)
Rotamer outliers (%)	3.44	1.99	3.96	2.94	1.5	5.37
Cβ outliers (%)	NA	NA	NA	NA	NA	NA
Peptide plane (%)						
Cis proline/general	0.0/0.0	0.0/0.0	0.0/0.0	0.0/0.0	0.0/0.0	0.0/0.0
Twisted proline/general	0.0/0.0	0.0/0.0	0.0/0.0	0.0/0.0	0.0/0.0	0.0/0.0
CaBLAM outliers (%)	1.85	1.74	0.9	1.2	1.01	1.42
ADP (B-factors)						
Iso/Aniso (#)	11117/0	10905/0	10585/0	10557/0	11103/0	11477/0
min/max/mean						

Supplementary Table 1. Collection and model statistics for the Cryo-EM structures.

Supplementary Table 1. (continued)

Protein	0.00/41.12/5.00	0.28/93.40/15.8 0	0.23/30.35/4.02	0/89.15/13.38	1.75/82.28/22.40	2.38/89.91/24.49
Nucleotide	0.23/83.07/30.5 6	3.71/150.32/64. 25	0.00/88.15/40.3 3	0.06/161.03/73. 80	4.85/140.55/59.47	6.08/156.41/67.10
Ligand	---	---	---	---	---	---
Water	---	---	---	---	---	---
Occupancy						
Mean	1	1	1	1	1	1
occ = 1 (%)	100	100	100	100	100	100
0 < occ < 1 (%)	0	0	0	0	0	0
occ > 1 (%)	0	0	0	0	0	0
Box						
Lengths (Å)	79.12, 119.54, 119.54	78.26, 118.68, 118.68	77.4, 116.96, 117.82	77.4, 116.96, 117.82	75.68, 116.1, 122.98	73.96, 116.1, 121.26
Angles (°)	90, 90, 90	90, 90, 90	90, 90, 90	90, 90, 90	90, 90, 90	90, 90, 90
Supplied Resolution (Å)	3.3	3.3	3	3	3.2	3.3
Resolution Estimates (Å)	Masked Unmasked	Masked Unmasked	Masked Unmasked	Masked Unmasked	Masked Unmasked	Masked Unmasked
d FSC (half maps; 0.143)	--- ---	--- ---	--- ---	--- ---	--- ---	--- ---
d 99 (full/half1/half2)	3.5/---/--- 3.5/---/---	3.5/---/--- 3.5/---/---	3.2/---/--- 3.2/---/---	3.2/---/--- 3.2/---/---	3.3/---/--- 3.3/--- /---	3.4/---/--- 3.3/--- /---
d model	3.4 3.4	3.4 3.4	3.1 3.2	3.1 3.2	3.2 3.2	3.3 3.3
d FSC model (0/0.143/0.5)	2.9/3.1/3.3 3.1/3.2/3.3	2.9/3.1/3.3 3.0/3.2/3.3	2.6/2.9/3.0 2.8/3.0/3.1	2.6/2.9/3.0 2.8/3.0/3.1	2.9/3.0/3.2 2.9/3.1/3.3	3.1/3.1/3.3 3.1/3.2/3.4
Map min/max/mean	39.16/62.30/0.7 9	- 39.16/62.3/0.82	41.24/58.89/0.4 0	41.24/58.89/0.4 0	-34.81/61.05/0.68	-31.70/51.07/0.56
Model vs Data						
CC (mask)	0.86	0.88	0.83	0.83	0.82	0.8
CC (box)	0.78	0.78	0.7	0.68	0.74	0.7
CC (peaks)	0.75	0.75	0.68	0.66	0.7	0.65
CC (volume)	0.8	0.82	0.75	0.76	0.78	0.76
Mean CC for ligands	---	---	---	---	---	---
PDB ID	90GS	90GZ	90GR	90H0	90H1	90H2

Supplementary Table 2. Alignment statistics of CnT-BS sequencing libraries.

XTC reads were downsampled to 30 million reads to match sperm samples, then all raw reads were trimmed and deduplicated using fastp v0.24.0 and aligned to the *Xenopus laevis* genome (Xenla 10.1) or lambda genome (to assess for bisulfite conversion efficiency) using Bismark v0.24.2. Methylated CpG percentages for H2A.Z samples were determined for the top 2.5 % of H2A.Z peaks by AUC (obtained using SEACR v1.3) while methylated percentages for H3 associated CpGs was conducted on regions outside of determined H2A.Z peaks. All CpGs met a cutoff of at least 5 reads to be counted. Methylated lambda CpG statistics were calculated from all mapped CpGs with no filtering.

		Alignment to <i>Xenopus laevis</i>			Alignment to lambda	
		Aligned %	Total Aligned Reads (No.)	Methylated CpG %	Aligned %	Methylated CpG %
Sperm Pronuclei						
H3	Rep 1	71.7	24141516	85.6	0.8	0.6
	Rep 2	71.0	24185147	85	1.0	0.6
H2AZ	Rep 1	64.7	25362485	42.5	4.0	0.6
	Rep 2	69.0	21963635	43.5	2.5	0.6
XTC-2						
H3	Rep 1	68.2	19539297	53.2	7.2	0.4
	Rep 2	62.4	17758471	53.8	15.9	0.3
H2AZ	Rep 1	69.1	19409396	3.2	12.4	0.3
	Rep 2	70.9	20109208	2.8	9.8	0.3

		Fragments in Bins (No.)	Fragments in Bins (%)
Sperm Pronuclei			
H3	<i>Rep 1</i>	9062125	34.04
	<i>Rep 2</i>	9265292	34.68
H2AZ	<i>Rep 1</i>	11482371	43.04
	<i>Rep 2</i>	9213698	39.58
XTC-2			
H3	<i>Rep 1</i>	7789646	36.43
	<i>Rep 2</i>	8145900	41.79
H2AZ	<i>Rep 1</i>	14276029	66.62
	<i>Rep 2</i>	15187573	69.13
Total Bins (No.)		506009	

Supplementary Table 3. Fragment statistics of filtered genomic bins used for sequencing analysis.

A count matrix was generated from processed bam files over 1,000 bp bins of the *Xenopus laevis* genome. Mitochondrial reads and bins containing fewer than 17 reads across less than 2 samples were removed to filter out low signal areas. Replicates for each sample were averaged and resulting count matrix used for genomic annotation analysis.

Data availability

DNA sequencing data can be accessed at GEO (accession number GSE295057) Raw image data and analysis code can be accessed at Zenodo DOI: 10.5281/zenodo.15360677 Cryo-EM maps and atomic models can be accessed with PDB IDs: 9OGS, 9OGZ (UM Sat2R-P); 9OGR, 9OH0 (Me Sat2R-P); 9OH1 (UM 601L); 9OH2 (Me 601L) and EMD IDs: EMD-70474 (UM Sat2R-P); EMD-70473 (Me Sat2R-P); EMD-70480 (UM 601L); EMD-70481 (Me 601L).

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Additional information

Author contributions

R.M.S. and H.F. conceived and designed the study. R.M.S. and Y.A. designed and performed the Cryo-EM analysis. R.M.S. and H.A.K. designed physiological experiments. H.A.K. created the custom α -SRCAP antibody, prepared XTC-2 nuclei, and purified pG-Tn5. R.M.S. conducted all experiments and analysis. R.M.S. and H.F. prepared the manuscript with input from H.A.K. and Y.A.

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Author ORCID iDs

Rochelle M Shih: <https://orcid.org/0000-0002-3218-4479>

Yasuhiro Arimura: <https://orcid.org/0000-0002-1903-6076>

Hide A Konishi: <https://orcid.org/0000-0001-9529-9321>

Hironori Funabiki: <https://orcid.org/0000-0003-4831-4087>

Additional files

[Supplementary Data 1](#) 

[Supplementary Data 2](#) 

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Peer reviews

Reviewer #1 (Public review):

Summary:

The authors considered the mechanism underlying previous observations that H2A.Z is preferentially excluded from methylated DNA regions. They considered two non-mutually exclusive mechanisms. First, they tested the hypothesis that nucleosomes containing both methylated DNA and H2A.Z might be intrinsically unstable due to their structural features. Second, they explored the possibility that DNA methylation might impede SRCAP-C from efficiently depositing H2A.Z onto these DNA methylated regions.

Their structural analyses revealed subtle differences between H2A.Z-containing nucleosomes assembled on methylated versus unmethylated DNA. To test the second hypothesis, the authors allowed H2A.Z assembly on sperm chromatin in *Xenopus* egg extracts and mapped both H2A.Z localization and DNA methylation in this transcriptionally inactive system. They compared these data with corresponding maps from a transcriptionally active *Xenopus* fibroblast cell line. This comparison confirmed the preferential deposition or enrichment of H2A.Z on unmethylated DNA regions, an effect that was much more pronounced in the fibroblast genome than in sperm chromatin. Furthermore, nucleosome assembly on methylated versus unmethylated DNA, along with SRCAP-C depletion from *Xenopus* egg extracts, provided a means to test whether SRCAP-C contributes to the preferential loading of H2A.Z onto unmethylated DNA.

Strengths:

The strength and originality of this work lie in its focused attempt to dissect the unexplained observation that H2A.Z is excluded from methylated genomic regions.

Weaknesses:

The study has two weaknesses. First, although the authors identify specific structural effects of DNA methylation on H2A.Z-containing nucleosomes, they do not provide evidence demonstrating that these structural differences lead to altered histone dynamics or nucleosome instability. Second, building on the elegant work of Berta and colleagues (cited in the manuscript), the authors implicate SRCAP-C in the selective deposition of H2A.Z at unmethylated regions. Yet the role of SRCAP-C appears only partial, and the study does not address how the structural or molecular consequences of DNA methylation prevent efficient H2A.Z deposition. Finally, additional plausible mechanisms beyond the two scenarios the authors considered are not investigated or discussed in the manuscript.

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Reviewer #2 (Public review):

This manuscript aims to elucidate the mechanistic basis for the long-standing observation that DNA methylation and the histone variant H2A.Z occupy mutually exclusive genomic regions. The authors test two hypotheses: (i) that DNA methylation intrinsically destabilizes H2A.Z nucleosomes, thereby preventing H2A.Z retention, and (ii) that DNA methylation suppresses H2A.Z deposition by ATP-dependent chromatin-remodelling complexes. However, neither hypothesis is rigorously addressed. There are experimental caveats, issues with data interpretation, and conclusions that are not supported by the data. Substantial revision and additional experiments, including controls, would be required before mechanistic conclusions can be drawn. Major concerns are as follows:

(1) The cryo-EM structure of methylated H2A.Z nucleosomes is insufficiently resolved to address the central mechanistic question: where the methylated CpGs are located relative to DNA-histone contact points and how these modifications influence H2A.Z nucleosome structure. The structure provides no mechanistic insights into methylation-induced destabilization.

The experimental system also lacks physiological relevance. The template DNA sequence is artificial, despite the existence of well-characterised native genomic sequences for which DNA methylation is known to inhibit H2A.Z incorporation. Alternatively, there are a number of studies examining the effect of DNA methylation on nucleosome structure, stability, DNA unwrapping, and positioning. Choosing one of these DNA sequences would have at least allowed a direct comparison with a canonical nucleosome. Indeed, a major omission is the absence of a cryo-EM structure of a canonical nucleosome assembled on the same DNA template - this is essential to assess whether the observed effects are H2A.Z-specific.

Furthermore, the DNA template is methylated at numerous random CpG sites. The authors' argument that only the global methylation level is relevant is inconsistent with the literature, which clearly demonstrates that methylation effects on canonical nucleosomes are position-dependent. Not all CpG sites contribute equally to nucleosome stability or unwrapping, and this critical factor is not considered.

Finally, and most importantly, the reported increase in accessibility of the methylated H2A.Z nucleosome is negligible compared with the much larger intrinsic DNA accessibility of the unmethylated H2A.Z nucleosome. These data do not support the authors' hypothesis and contradict the manuscript's conclusions. Claims that methylated H2A.Z nucleosomes are "more open and accessible" must therefore be removed, and the title is misleading, given that no meaningful impact of DNA methylation on H2A.Z nucleosome stability is demonstrated.

(2) The cryo-EM structures of methylated and unmethylated 601L H2A.Z nucleosomes show no detectable differences. As presented, this negative result adds little value. If anything, it

reinforces the point that the positional context of CpG methylation is critical, which the manuscript does not consider.

(3) Very little H3 signal coincides with H2A.Z at TSSs in sperm pronuclei, yet this is neither explained nor discussed (Supplementary Figure 10D). The authors need to clarify this.

(4) In my view, the most conceptually important finding is that H2A.Z-associated reads in sperm pronuclei show ~43% CpG methylation. This directly contradicts the model of strict mutual exclusivity and suggests that the antagonism is context-dependent. Similarly, the finding that the depletion of SRCAP reduces H2A.Z deposition only on unmethylated templates is also very intriguing. Collectively, these results warrant further investigation (see below).

(5) Given that H2A.Z is located at diverse genomic elements (e.g., enhancers, repressed gene bodies, promoters), the manuscript requires a more rigorous genomic annotation comparing H2A.Z occupancy in sperm pronuclei versus XTC-2 cells. The authors should stratify H2A.Z-DNA methylation relationships across promoters, 5'UTRs, exons, gene bodies, enhancers, etc., as described in Supplementary Figure 10A.

(6) Although H2A.Z accumulates less efficiently on exogenous methylated substrates in egg extract, substantial deposition still occurs (~50%). This observation directly challenges the strong antagonistic model described in the manuscript, yet the authors do not acknowledge or discuss it. Moreover, differences between unmethylated and methylated 601 DNA raise further questions about the biological relevance of the cryo-EM 601 structures.

(7) The SRCAP depletion is insufficiently validated i.e., the antibody-mediated depletion of SRCAP lacks quantitative verification. A minimum of three biological replicates with quantification is required to substantiate the claims.

(8) It appears that the role of p400-Tip60 has been completely overlooked. This complex is the second major H2A.Z deposition complex. Because p400 exhibits DNA methylation-insensitive binding (Supplementary Figure 14), it may account for the deposition of H2A.Z onto methylated DNA. This possibility is highly significant and must be addressed by repeating the key experiments in Figure 5 following p400-Tip60 depletion.

(9) The manuscript repeatedly states that H2A.Z nucleosomes are intrinsically unstable; however, this is an oversimplification. Although some DNA unwrapping is observed, multiple studies show that H3/H4 tetramer-H2A.Z/H2B interactions are more stable (important recent studies include the following: DOI: 10.1038/s41594-021-00589-3; 10.1038/s41467-021-22688-x; and reviewed in 10.1038/s41576-024-00759-1).

In summary, the current manuscript does not present a convincing mechanistic explanation for the antagonism between DNA methylation and H2A.Z. The observation that H2A.Z can substantially coexist with DNA methylation in sperm pronuclei, perhaps, should be the conceptual focus.

<https://doi.org/10.7554/eLife.109762.1.sa2>

Reviewer #3 (Public review):

Summary:

Histone variant H2A.Z is evolutionarily conserved among various species. The selective incorporation and removal of histone variants on the genome play crucial roles in regulating nuclear events, including transcription. Shih et al. aimed to address antagonistic mechanisms between histone variant H2A.Z deposition and DNA methylation. To this end, the authors reconstituted H2A.Z nucleosomes in vitro using methylated or unmethylated human satellite

II DNA sequence and examined how DNA methylation affects H2A.Z nucleosome structure and dynamics. The cryo-EM analysis revealed that DNA methylation induces a more open conformation in H2A.Z nucleosomes. Consistent with this, their biochemical assays showed that DNA methylation subtly increases restriction enzyme accessibility in H2A.Z nucleosomes compared with canonical H2A nucleosomes. The authors identified genome-wide profiles of H2A.Z and DNA methylation using genomic assays and found their unique distribution between *Xenopus* sperm pronuclei and fibroblast cells. Using *Xenopus* egg extract systems, the authors showed SRCAP complex, the chromatin remodelers for H2A.Z deposition, preferentially deposit H2A.Z on unmethylated DNA.

Strengths:

The study is solid, and most conclusions are well-supported. The experiments are rigorously performed, and interpretations are clear. The study presents a high-resolution cryo-EM structure of human H2A.Z nucleosome with methylated DNA. The discovery that the SRCAP complex senses DNA methylation is novel and provides important mechanistic insight into the antagonism between H2A.Z and DNA methylation.

Weaknesses:

The study is already strong, and most conclusions are well supported. However, it can be further strengthened in several ways.

- (1) It is difficult to interpret how DNA methylation alters the orientation of the H4 tail and leads to the additional density on the acidic patch. The data do not convincingly support whether DNA methylation enhances interactions with H2A.Z mono-nucleosomes, nor whether this effect is specific to methylated H2A.Z nucleosomes.
- (2) It remains unclear whether DNA methylation alters global H2A.Z nucleosome stability or primarily affects local DNA end flexibility. Moreover, while the authors showed locus-specific accessibility by *HinfI* digestion, an unbiased assay such as MNase digestion would strengthen the conclusions.

<https://doi.org/10.7554/eLife.109762.1.sa1>

Author response:

eLife Assessment

This study provides valuable mechanistic insight into the mutually exclusive distributions of the histone variant H2A.Z and DNA methylation by testing two hypotheses: (i) that DNA methylation destabilizes H2A.Z nucleosomes, thereby preventing H2A.Z retention, and (ii) that DNA methylation suppresses H2A.Z deposition by ATP-dependent chromatin remodeling complexes. Through a series of well-designed and carefully executed experiments, findings are presented in support of both hypotheses. However, the evidence in support of either hypothesis is incomplete, so that the proposed mechanisms underlying the enrichment of H2A.Z on unmethylated DNA remain somewhat speculative.

We would like to thank the editor and reviewers for their critical assessments of our manuscript. While we do acknowledge the limitations of our work, we believe that our results provide important mechanistic insights into the long-standing question of how H2A.Z is preferentially enriched in hypomethylated genomic DNA regions. First, our structural and biochemical data suggest that DNA methylation increases the openness and physical accessibility of H2A.Z, albeit the effect is relatively subtle and is sequence-dependent. Second, using *Xenopus* egg extracts and synthetic DNA templates, we provide the first clear and direct evidence that DNA methylation-sensitive H2A.Z deposition is due to the H2A.Z chaperone SRCAP-C, corroborated by our discovery that SRCAP-C binding to DNA is suppressed by DNA

methylation. Although the molecular details by which DNA methylation inhibits binding of SRCAP-C is an important area of future study, in our current manuscript, we do provide evidence that directly links the presence of SRCAP-C to the establishment of the DNA methylation/H2A.Z antagonism in a physiological system. Thanks to criticisms by the reviewers, we realized that we did not clearly state in our Abstract that the impact of DNA methylation on intrinsic H2A.Z nucleosome stability is relatively subtle, although we did explain these observations and limitations in the main text. In our revised manuscript, we are willing to edit the text to better clarify the criticisms raised by the reviewers.

Public Reviews:

Reviewer #1 (Public review):

Summary:

The authors considered the mechanism underlying previous observations that H2A.Z is preferentially excluded from methylated DNA regions. They considered two non-mutually exclusive mechanisms. First, they tested the hypothesis that nucleosomes containing both methylated DNA and H2A.Z might be intrinsically unstable due to their structural features. Second, they explored the possibility that DNA methylation might impede SRCAP-C from efficiently depositing H2A.Z onto these DNA methylated regions.

*Their structural analyses revealed subtle differences between H2A.Z-containing nucleosomes assembled on methylated versus unmethylated DNA. To test the second hypothesis, the authors allowed H2A.Z assembly on sperm chromatin in *Xenopus* egg extracts and mapped both H2A.Z localization and DNA methylation in this transcriptionally inactive system. They compared these data with corresponding maps from a transcriptionally active *Xenopus* fibroblast cell line. This comparison confirmed the preferential deposition or enrichment of H2A.Z on unmethylated DNA regions, an effect that was much more pronounced in the fibroblast genome than in sperm chromatin. Furthermore, nucleosome assembly on methylated versus unmethylated DNA, along with SRCAP-C depletion from *Xenopus* egg extracts, provided a means to test whether SRCAP-C contributes to the preferential loading of H2A.Z onto unmethylated DNA.*

Strengths:

The strength and originality of this work lie in its focused attempt to dissect the unexplained observation that H2A.Z is excluded from methylated genomic regions.

Weaknesses:

The study has two weaknesses. First, although the authors identify specific structural effects of DNA methylation on H2A.Z-containing nucleosomes, they do not provide evidence demonstrating that these structural differences lead to altered histone dynamics or nucleosome instability. Second, building on the elegant work of Berta and colleagues (cited in the manuscript), the authors implicate SRCAP-C in the selective deposition of H2A.Z at unmethylated regions. Yet the role of SRCAP-C appears only partial, and the study does not address how the structural or molecular consequences of DNA methylation prevent efficient H2A.Z deposition. Finally, additional plausible mechanisms beyond the two scenarios the authors considered are not investigated or discussed in the manuscript.

Although we acknowledge the limitations of our study and are willing to expand our discussion to more thoroughly discuss these points, we believe our manuscript provides several important mechanistic insights which this reviewer may not have fully appreciated.

Our first conclusion that H2A.Z nucleosomes on methylated DNA are more open and accessible compared to their unmethylated counterparts is supported by both our cryo-EM study and the restriction enzyme accessibility assay. Although the physical effect of DNA methylation is relatively subtle and is likely sequence dependent, as we clearly noted within the manuscript, the difference does exist and is valuable information for the chromatin field at large to consider.

The second major conclusion of our manuscript is that SRCAP-C exhibits preferential binding to unmethylated DNA over methylated DNA, and that SRCAP-C represents the major mechanism that can explain the biased deposition of H2A.Z to unmethylated DNA in *Xenopus* egg extracts. Furthermore, our experiments using *Xenopus* egg extract clearly demonstrated that H2A.Z is deposited by both DNA-methylation sensitive and insensitive mechanisms. Depletion of SRCAP-C almost completely eliminated the levels of DNA-methylation-sensitive H2A.Z deposition and reduced the total level of H2A.Z on chromatin to less than half of that seen in non-depleted extract. This result demonstrated that DNA methylation-sensitive H2A.Z loading is primarily regulated by SRCAP-C, at least in our experimental context where transcription, replication, and other epigenetic modifications are not involved. It is likely that additional mechanisms do further contribute, implicated by our sequencing experiments, particularly at regions with active transcription, and we have noted these possibilities and the rationale for their existence in the Discussion.

Our study also suggests that a SRCAP-independent, DNA methylation-insensitive mechanism of H2A.Z loading exists, which we suspect to be mediated by Tip60-C. In line with this possibility, our data suggest that Tip60-C binds DNA in a DNA methylation-insensitive manner in *Xenopus* egg extract. Since antibodies to deplete Tip60-C from *Xenopus* egg extract are currently unavailable, we were unable to directly test that hypothesis and decided not to include Tip60-C into our final model as we lacked experimental evidence for its role. However, whether or not Tip60-C is the complex responsible for the DNA methylation-insensitive pathway does not influence our final conclusion that SRCAP-C plays a major role in DNA methylation-sensitive H2A.Z loading. We are planning to edit our manuscript to more comprehensively discuss these points.

Please note that while Berta et al reported that DNA methylation increases at H2A.Z loci in tumors defective in SRCAP-C, they selected those regions based off where H2A.Z is typically enriched within normal tissues (Berta et al., 2021). They did not show data indicating whether H2A.Z is still retained specifically at those analyzed loci upon mutation of SRCAP-C subunits. Thus, although we greatly admire their work and are pleased that many of our findings align with theirs, their paper did not directly address whether SRCAP-C itself differentiates between DNA methylation status nor the impact that has on H2A.Z and DNA methylation colocalization. In contrast, our *Xenopus* egg extract system, where de novo methylation is undetectable (Nishiyama et al., 2013; Wassing et al., 2024) offers a unique opportunity to examine the direct impact of DNA methylation on H2A.Z deposition using controlled synthetic DNA substrates. Corroborated with our demonstration that DNA binding of SRCAP-C is suppressed by DNA methylation, we believe that our manuscript provides a specific mechanism that can explain the preferential deposition of H2A.Z at hypomethylated genomic regions.

Reviewer #2 (Public review):

This manuscript aims to elucidate the mechanistic basis for the long-standing observation that DNA methylation and the histone variant H2A.Z occupy mutually exclusive genomic regions. The authors test two hypotheses: (i) that DNA methylation intrinsically destabilizes H2A.Z nucleosomes, thereby preventing H2A.Z retention, and (ii) that DNA methylation suppresses H2A.Z deposition by ATP-dependent chromatin-remodelling complexes. However, neither hypothesis is rigorously addressed. There are

experimental caveats, issues with data interpretation, and conclusions that are not supported by the data. Substantial revision and additional experiments, including controls, would be required before mechanistic conclusions can be drawn. Major concerns are as follows:

We appreciate the critical assessment of our manuscript by this reviewer. Although we acknowledge the limitations of our study and will revise the manuscript to better describe them, we would like to respectfully argue against the statement that our "conclusions [...] are not supported by the data".

(1) The cryo-EM structure of methylated H2A.Z nucleosomes is insufficiently resolved to address the central mechanistic question: where the methylated CpGs are located relative to DNA-histone contact points and how these modifications influence H2A.Z nucleosome structure. The structure provides no mechanistic insights into methylation-induced destabilization.

The fact that the DNA resolution in the methylated structure was not high enough to resolve the positions of methylated CpGs despite a high overall resolution of 2.78 Å implies that 1) the Sat2R-P DNA was not as stably registered as the 601L sequence, requiring us to create two alternative Sat2R-P atomic models to account for the variable positioning in our samples, and 2) that the presence of DNA methylation increases that positional variability. We understand that one may prefer to see highly resolved density around each methylation mark, but we do believe that our inability to accomplish that is actually a feature rather than a weakness and has important biological implications. The decrease in local DNA resolution on the methylated Sat2R-P structure compared to its unmethylated counterpart is meaningful and suggests to us that DNA methylation weakens overall DNA wrapping and positioning on the nucleosome, supported by the increased flexibility seen at the linker DNA ends as well as an increase in the population of highly shifted nucleosomes amongst the methylated particles. Additionally, one major view in the DNA methylation/nucleosome stability field is that the presence of DNA methylation can make DNA stiffer and harder to bend, causing opening and destabilization of nucleosomes (Ngo et al., 2016). The increased opening of linker DNA ends and accessibility of methylated H2A.Z nucleosomes in our hands also aligns with such an idea, again suggesting decreased histone-DNA contact stability on methylated DNA substrates. We plan to revise the writing in our manuscript to better reflect these ideas.

The experimental system also lacks physiological relevance. The template DNA sequence is artificial, despite the existence of well-characterised native genomic sequences for which DNA methylation is known to inhibit H2A.Z incorporation. Alternatively, there are a number of studies examining the effect of DNA methylation on nucleosome structure, stability, DNA unwrapping, and positioning. Choosing one of these DNA sequences would have at least allowed a direct comparison with a canonical nucleosome. Indeed, a major omission is the absence of a cryo-EM structure of a canonical nucleosome assembled on the same DNA template - this is essential to assess whether the observed effects are H2A.Z-specific.

The reviewer raises a fair question about whether canonical H2A would experience the same DNA methylation-dependent structural effects. We had considered solving the H2A structures, however, ultimately decided against it for a few reasons. First, there already exists crystal structures of canonical H2A nucleosomes using a DNA sequence highly similar to our Sat2R-P with and without the presence of DNA methylation (PDB: 5CPI and 5CPJ). The authors of this study did not see any physical differences present in their structures (Osakabe et al., 2015). Additionally, we had included canonical H2A conditions within our restriction enzyme accessibility assay and did not see a significant impact of DNA methylation on those samples (Fig 3). Because of the previous report and our own negative data, we expected that only limited additional insights would be obtained from the canonical H2A structures and decided not to pursue that analysis.

One of the primary reasons we chose the Sat2R-P sequence was, as noted above, that there already was a published study examining how DNA methylation affects nucleosome structure using a variant of this sequence which we could compare to our results, as the reviewer has suggested. We did have to modify the sequence, namely by making it palindromic, in order to increase the final achievable resolution. We viewed the Sat2R-P sequence as an attractive candidate because it is physiologically relevant; the initial sequence was taken directly from human satellite II. Several modifications were made for technical reasons, including making the sequence palindromic as described above and also ensuring that each CpG is recognizable by a methylation-sensitive restriction enzyme so that we could be certain about the degree of methylation on our substrates. These practical concerns outweighed the necessity of maintaining a strict physiological sequence to us. However, we still believe the final Sat2R-P more closely mimics physiological sequences than Widom 601. Additionally, human satellite II is a highly abundant sequence in the human genome that is known to undergo large methylation changes on the onset of many disorders, like cancer, as well as during aging. Thus, there are interesting biological questions surrounding how the methylation state of this particular sequence affects chromatin structure. Furthermore, it has been reported that satellite II is devoid of H2A.Z (Capurso et al., 2012). Beyond those reasons, the satellite II sequence is generally interesting to our lab because we have been studying genes involved in ICF syndrome, where hypomethylation of satellite II sequences forms one of the hallmarks of this disorder (Funabiki et al., 2023; Jenness et al., 2018; Wassing et al., 2024). We understand that sequence context plays a large role in nucleosome wrapping and stability. This is why we strived to test multiple sequences in each of our assays. We do agree that it would be interesting to use DNA sequences where H2A.Z binding has already been described to be affected in a DNA methylation-dependent manner, forming an exciting future study to pursue.

Furthermore, the DNA template is methylated at numerous random CpG sites. The authors' argument that only the global methylation level is relevant is inconsistent with the literature, which clearly demonstrates that methylation effects on canonical nucleosomes are position-dependent. Not all CpG sites contribute equally to nucleosome stability or unwrapping, and this critical factor is not considered.

We did not argue that only the global methylation level is relevant. We also would appreciate it if the reviewer could provide specific references that "clearly demonstrates that methylation effects on canonical nucleosomes are position-dependent". We are aware of a series of studies conducted by Chongli Yuan's group, including one testing the effect of placing methylated CpGs at different positions along the Widom 601 sequence. In that study (Jimenez-Useche et al., 2013), they did find that positioning of mCpGs has differential impacts on the salt resistance of the nucleosomes, with 5 tandem mCpG copies at the dyad causing the most dramatic nucleosome opening whereas having mCpGs only at the DNA major grooves, but not elsewhere, increased nucleosome stability. However, they did also find that methylation of the original Widom 601 sequence also caused destabilization, albeit to a lesser degree, and another study by the same group (Jimenez-Useche et al., 2014) also found that CpG methylation decreased nucleosome-forming ability for all tested variants of the Widom 601 sequence, regardless of CpG density or positioning.

Other studies monitored how distribution of methylated CpGs correlates with nucleosome positioning (Collings et al., 2013; Davey et al., 1997; Davey et al., 2004). However, these studies assessed the sequence-dependent effects specifically on nucleosome assembly during in vitro salt dialysis, which is a different physical process than the one our manuscript focuses on, especially when considering the fact that H2A.Z is deposited onto preassembled H2A-nucleosome. Our cryo-EM analysis examines the structural changes induced by DNA methylation on already formed nucleosomes rather than the process of formation. Thus,

probing accessibility changes using a restriction enzyme was the more appropriate biochemical assay to verify our structures.

We do very much agree that DNA context can influence nucleosome stability under different conditions. A study of molecular dynamics simulations concluded that the "combination of overall DNA geometrical and shape properties upon methylation" makes nucleosomes resistant to unwrapping (Li et al., 2022), while another modeling study suggests that DNA methylation impacts nucleosome stability in a manner dependent on DNA sequence, where "[s]trong binding is weakened and weak binding is strengthened" (Minary and Levitt, 2014). While G/C-dinucleotides are preferentially placed at major groove-inward positions in the nucleosomes in vivo (Chodavarapu et al., 2010; Segal et al., 2006) and G/C-rich segments are excluded from major groove-outward positions in Widom 601-like nucleosomes (Chua et al., 2012), methylated CpG dinucleotides are preferably, if not exclusively, located at major groove-outward positions in vivo. Mechanisms behind this biased mCpG positioning on the nucleosome remain speculative, likely caused by a combination of multiple factors, but the fact that we did not observe clear structural impacts using the Widom 601L sequence, where mCpGs are located at the major groove-outward and -inward positions ((Chua et al., 2012) and our structure), deserves a space for discussion. On the other hand, positioning of mCpG on satellite II-derived sequences that we used in this study was based on a physiological sequence, and thus it may not be appropriate to say that those CpGs are placed at multiple "random" positions. Although we decided not to discuss the position of 5mC on our Sat2R nucleosome structure due to ambiguous base assignments, neither of our two atomic models is consistent with an idea that DNA methylation repositions the CpG to the outward major grooves. As the potential contribution of how DNA methylation affects the nucleosome structure via modulating DNA stiffness has been extensively studied (Choy et al., 2010; Li et al., 2022; Ngo et al., 2016; Perez et al., 2012), we believe that it is appropriate to consider overall DNA properties along the whole DNA sequence, though we are willing to discuss potential positional effects in the revised manuscript.

Perhaps one of the most important points that we did not emphasize enough in our original manuscript was that in contrast to the subtle intrinsic effect of DNA methylation that was DNA sequence dependent, we observed SRCAP-dependent preferential H2A.Z deposition to unmethylated DNA over methylated DNA in both 601 and satellite II DNAs. In the revised manuscript, we will make the value of comparative studies on 601 and satellite II in two distinct mechanisms.

Finally, and most importantly, the reported increase in accessibility of the methylated H2A.Z nucleosome is negligible compared with the much larger intrinsic DNA accessibility of the unmethylated H2A.Z nucleosome. These data do not support the authors' hypothesis and contradict the manuscript's conclusions. Claims that methylated H2A.Z nucleosomes are "more open and accessible" must therefore be removed, and the title is misleading, given that no meaningful impact of DNA methylation on H2A.Z nucleosome stability is demonstrated.

We respectfully disagree with this reviewer's criticism. We investigated the potential impact of DNA methylation on nucleosome stability to the best of our abilities through complementary assays and reported our observations. The effect of DNA methylation is smaller than the difference between H2A.Z and H2A, but we were able to see an effect. It is also not uncommon for small differences to have functional impacts in biological systems. We agree that further testing is required to determine whether this subtle effect is functionally important, and it remains the subject of future research due to the many technical challenges associated with addressing said question. We would like to note that 18 years have passed since Daniel Zilberman first reported the antagonistic relationship between H2AZ and DNA methylation (Zilberman et al., 2008) but very few studies have since directly tested specific mechanistic hypotheses. We believe that our study lays the groundwork for exciting future

investigation that better elucidates the pathways that contribute to this antagonism and will have meaningful impacts on the field in general. However, thanks to the reviewer's criticism, we realized that we did not clearly state in the Abstract the relatively subtle effect of DNA methylation on the intrinsic H2A.Z nucleosome stability. Therefore, we will accordingly revise the Abstract to make this point clearer.

(2) The cryo-EM structures of methylated and unmethylated 601L H2A.Z nucleosomes show no detectable differences. As presented, this negative result adds little value. If anything, it reinforces the point that the positional context of CpG methylation is critical, which the manuscript does not consider.

We believe the inclusion and factual reporting of negative data is important for the scientific community as one of the major issues currently in biology research is biased omission of negative data. We considered eLife as a venue to publish this work for this reason. We understand that the reviewer believes our 601L structures may detract from the overall message of our manuscript. We believe this data rather emphasizes the importance of DNA sequence context, something that the reviewer also rightfully notes. It is standard practice in the nucleosome field to use the Widom 601 sequence, along with its variants. Our experience has shown that use of an artificially strong positioning sequence may mask weaker physical effects that could play a physiological role. Thus, we were careful to validate all further assays with multiple DNA sequences and believed it important to report these sequence-dependent effects on nucleosome structure.

(3) Very little H3 signal coincides with H2A.Z at TSSs in sperm pronuclei, yet this is neither explained nor discussed (Supplementary Figure 10D). The authors need to clarify this.

Our H3 signal, which represents the global nucleosome population, is more broadly distributed across the genome than H2A.Z, which is known to localize at specific genomic sites. Since both histone types were sequenced to similar read depths, H3 peaks are generally shallower than H2A.Z and peak heights cannot be directly compared (i.e. they should be represented in separate appropriate data ranges).

(4) In my view, the most conceptually important finding is that H2A.Z-associated reads in sperm pronuclei show ~43% CpG methylation. This directly contradicts the model of strict mutual exclusivity and suggests that the antagonism is context-dependent. Similarly, the finding that the depletion of SRCAP reduces H2A.Z deposition only on unmethylated templates is also very intriguing. Collectively, these result warrants further investigation (see below).

(5) Given that H2A.Z is located at diverse genomic elements (e.g., enhancers, repressed gene bodies, promoters), the manuscript requires a more rigorous genomic annotation comparing H2A.Z occupancy in sperm pronuclei versus XTC-2 cells. The authors should stratify H2A.Z-DNA methylation relationships across promoters, 5'UTRs, exons, gene bodies, enhancers, etc., as described in Supplementary Figure 10A.

(below is response to (4) and (5) together)

We agree that the substantial presence of co-localized H2A.Z and DNA methylation specifically in the sperm pronuclei samples and the changes in pattern between nuclear types are highly interesting and require further investigation. However, we faced technical challenges in our sequencing experiments that made us refrain from conducting a more detailed analysis for fear of over-interpreting potential artifacts. These challenges mainly stemmed from the difficulties in collecting enough material from *Xenopus* egg extracts and Tn5's innate bias towards accessible regions of the genome. Because of this, open regions of the genome tend to be overrepresented in our data (as noted in our Discussion), making it challenging to rigorously compare methylation profiles and H2A.Z/H3 associated genomic elements.

While the degree of separation seems to be dependent on nuclei type, we still believe the antagonism exists in both the sperm pronuclei and XTC-2 samples when comparing H2A.Z methylation profiles to the corresponding H3 condition. Our study also demonstrates that H2A.Z is preferentially deposited to hypomethylated DNA in a manner dependent of SRCAP-C (the loss of SRCAP only reduces H2A.Z on unmethylated substrates) but an additional methylation-insensitive H2A.Z deposition mechanism also exists. We realized that this interesting point was not clearly highlighted in Abstract, so we will revise it accordingly.

(6) Although H2A.Z accumulates less efficiently on exogenous methylated substrates in egg extract, substantial deposition still occurs (~50%). This observation directly challenges the strong antagonistic model described in the manuscript, yet the authors do not acknowledge or discuss it. Moreover, differences between unmethylated and methylated 601 DNA raise further questions about the biological relevance of the cryo-EM 601 structures.

As depicted in Figure 6 and described in the Discussion, we clearly indicated that both methylation-sensitive and methylation-insensitive pathways exist to deposit H2A.Z within the genome. We also directly stated in our Discussion that a substantial proportion of H2A.Z colocalizes with DNA methylation both in our study as well as in previous reports, which is of major interest for future study. Additionally, we further discussed how the absence of transcription in *Xenopus* eggs is a likely reason for the more limited effect of DNA methylation restricting H2A.Z deposition in our egg extract system.

As noted in our response to (2), the lack of a clear impact on our 601L structures implies that this is due to the extraordinarily strong artificial nucleosome positioning capacity of the 601 sequence and its variants. Since 601 is heavily used in chromatin biology, including within DNA methylation research, such negative data are still useful to include and publish.

(7) The SRCAP depletion is insufficiently validated i.e., the antibody-mediated depletion of SRCAP lacks quantitative verification. A minimum of three biological replicates with quantification is required to substantiate the claims.

We are willing to address this concern. However, please note that our data showed that methylation-dependent H2A.Z deposition is almost completely erased upon SRCAP depletion, indicating functionally effective depletion. The specificity of the custom antibody against *Xenopus* SRCAP was verified by mass spectrometry. Additionally, we have obtained the same effect using another commercially available SRCAP antibody, though we did not include this preliminary result in our original manuscript. Due to its relatively low abundance and high molecular weight, SRCAP western blot signals are weak, making it challenging to quantify the degree of depletion. We also believe that the value of quantification in this context, with the points noted above, is rather limited. In the past, our lab has published papers on depleting the H3T3 kinase Haspin from *Xenopus* egg extracts (Ghenoiu et al., 2013; Kelly et al., 2010) but were never able to detect Haspin via western blot. This protein was only detected by mass spectrometry specifically on nucleosome array beads with H3K9me3 (Jenness et al., 2018). However, depletion of Haspin was readily monitored by erasure of H3T3ph, the enzymatic product of Haspin. In these experiments, it was impossible, and not critical, to quantitatively monitor the depletion of Haspin protein in order to investigate its molecular functions. Similarly, in this current study, the important fact is that depletion of SRCAP suppressed methylation-sensitive H2A.Z deposition and quantifying the degree of SRCAP depletion would not have a major impact on this conclusion.

(8) It appears that the role of p400-Tip60 has been completely overlooked. This complex is the second major H2A.Z deposition complex. Because p400 exhibits DNA methylation-insensitive binding (Supplementary Figure 14), it may account for the deposition of H2A.Z

onto methylated DNA. This possibility is highly significant and must be addressed by repeating the key experiments in Figure 5 following p400-Tip60 depletion.

We are aware that the Tip60 complex is a very likely candidate for mediating DNA methylation-insensitive H2A.Z deposition, which is why we tested whether DNA binding of p400 is methylation sensitive. Therefore, the reviewer's statement that we "completely overlooked" Tip60-C's role does not fairly report on our efforts. We wished to test the potential contribution of Tip60-C, but, unfortunately, the antibodies we currently have available to us were not successful in depleting the complex from egg extract. Since we had no direct experimental evidence indicating the role Tip60-C plays, we decided to take a conservative approach to our model and leave the methylation-insensitive pathway as mediated by something still unidentified. While further investigating Tip60-C's contribution to this pathway is of definite value, we do not believe that it impacts our major conclusion that SRCAP-C is the main mediator responsible for H2A.Z deposition on unmethylated DNA and thus remains a subject for future study.

(9) The manuscript repeatedly states that H2A.Z nucleosomes are intrinsically unstable; however, this is an oversimplification. Although some DNA unwrapping is observed, multiple studies show that H3/H4 tetramer-H2A.Z/H2B interactions are more stable (important recent studies include the following: DOI: 10.1038/s41594-021-00589-3; 10.1038/s41467-021-22688-x; and reviewed in 10.1038/s41576-024-00759-1).

We understand that the H2A.Z stability field is highly controversial. We have introduced the many conflicting reports that have been published in the field but can further expand on the controversies if desired. We also understand that the term "nucleosome stability" is broad and encompasses many physical aspects. As noted in a prior response, we will better specify our use of the term within the manuscript. In our assays, we are most focused on the DNA wrapping stability of the nucleosome and have consistently seen in our hands that H2A.Z nucleosomes are much more open and accessible compared to canonical H2A on satellite II-derived sequences, regardless of methylation status. However, we do understand that many groups have observed the opposite findings while others have obtained results similar to us. We reported on our findings of the general H2A.Z stability with the hopes to help clarify some of the field's controversies.

In summary, the current manuscript does not present a convincing mechanistic explanation for the antagonism between DNA methylation and H2A.Z. The observation that H2A.Z can substantially coexist with DNA methylation in sperm pronuclei, perhaps, should be the conceptual focus.

We appreciate this reviewer's advice. However, please note that the first author who led this project has already successfully defended their PhD thesis primarily based on this project, making it impractical and unrealistic to completely change the focus of this manuscript to include an entirely new avenue of research. We believe that our data provide important insights into the mechanisms by which H2A.Z is excluded from methylated DNA, particularly via the DNA methylation-sensitive binding of SRCAP-C, which has never been described before. We agree that many questions are still left unanswered, including the exact molecular mechanism behind how DNA methylation prevents SRCAP-C binding. We have preliminary data that suggest none of the known DNA-binding modules of SRCAP-C, including ZNHIT1, by themselves can explain this sensitivity. This implies that domain dissection in the context of the holo-SRCAP complex is required to fully address this question. We believe this represents a very exciting future avenue of study; however, it does not negate our finding that SRCAP-C itself is important for maintaining the DNA methylation/H2A.Z antagonism. Therefore, we respectfully disagree with this reviewer's summary statement, which misleadingly undermines the impact of our work.

Reviewer #3 (Public review):

Summary:

*Histone variant H2A.Z is evolutionarily conserved among various species. The selective incorporation and removal of histone variants on the genome play crucial roles in regulating nuclear events, including transcription. Shih et al. aimed to address antagonistic mechanisms between histone variant H2A.Z deposition and DNA methylation. To this end, the authors reconstituted H2A.Z nucleosomes in vitro using methylated or unmethylated human satellite II DNA sequence and examined how DNA methylation affects H2A.Z nucleosome structure and dynamics. The cryo-EM analysis revealed that DNA methylation induces a more open conformation in H2A.Z nucleosomes. Consistent with this, their biochemical assays showed that DNA methylation subtly increases restriction enzyme accessibility in H2A.Z nucleosomes compared with canonical H2A nucleosomes. The authors identified genome-wide profiles of H2A.Z and DNA methylation using genomic assays and found their unique distribution between *Xenopus* sperm pronuclei and fibroblast cells. Using *Xenopus* egg extract systems, the authors showed SRCAP complex, the chromatin remodelers for H2A.Z deposition, preferentially deposit H2A.Z on unmethylated DNA.*

Strengths:

The study is solid, and most conclusions are well-supported. The experiments are rigorously performed, and interpretations are clear. The study presents a high-resolution cryo-EM structure of human H2A.Z nucleosome with methylated DNA. The discovery that the SRCAP complex senses DNA methylation is novel and provides important mechanistic insight into the antagonism between H2A.Z and DNA methylation.

We are grateful that this reviewer recognizes the importance of our study.

Weaknesses:

The study is already strong, and most conclusions are well supported. However, it can be further strengthened in several ways.

(1) It is difficult to interpret how DNA methylation alters the orientation of the H4 tail and leads to the additional density on the acidic patch. The data do not convincingly support whether DNA methylation enhances interactions with H2A.Z mono-nucleosomes, nor whether this effect is specific to methylated H2A.Z nucleosomes.

The altered H4 tail orientation and extra density seen on the acidic patch were incidental findings that we thought could be interesting for the field to be aware of but decided not to follow up on as there were other structural differences that were more directly related to our central question. We do believe that the above two differences are linked to each other because we used a highly purified and homogenous sample for cryo-EM analysis and the H4 tail/acidic patch interaction is a well characterized contact that mediates inter-nucleosome interactions. Additionally, other groups have reported that the presence of DNA methylation causes condensation of both chromatin and bare DNA (cited within our manuscript), though the mechanics behind this phenomenon remain to be elucidated. We believed that our structure data may also align with those findings. However, the reviewer is fair in pointing out that we do not provide further experimental evidence in verifying the existence of these increased interactions. We can revise our writing to clarify that these points are currently hypotheses rather than validated results.

*(2) It remains unclear whether DNA methylation alters global H2A.Z nucleosome stability or primarily affects local DNA end flexibility. Moreover, while the authors showed locus-specific accessibility by *HinfI* digestion, an unbiased assay such as MNase digestion would strengthen the conclusions.*

We would like to thank the reviewer for bringing up these issues. Although our current data cannot explicitly clarify these possibilities, we favor an idea that DNA methylation specifically alters histone to DNA contacts and that this effect is felt globally across the entire nucleosome rather than only at specific locations. The intrinsic flexibility of linker DNA ends means that that region tends to exhibit the greatest differences under different physical influences, hence the focus on characterizing that area; flexibility of a thread on a spool is most pronounced at the ends. However, we also found that the DNA backbone of H2A.Z on methylated DNA had a lower local resolution compared to its unmethylated counterpart, despite that structure having a higher global resolution, which suggested to us that DNA positioning along the nucleosome is overall weaker under the presence of DNA methylation. This is corroborated by the increased population of open/shifted structures in our classification analysis. The reviewer raises a fair point about the use of a specific restriction enzyme versus MNase. We agree that our accessibility assay is highly influenced by the position of the restriction site and have previously seen that moving the cut site too close to the linker DNA end will abolish any DNA methylation-dependent differences. We did initially attempt an MNase digestion-based assay, but the data were not as reproducible as with the use of a specific restriction enzyme. We do not know the reason behind this irreproducibility though we believe that the processivity of MNase could make it difficult to capture subtle effects like those induced by DNA methylation on already highly accessible H2A.Z nucleosomes. Overall, while we believe that DNA methylation does exert a physical effect, its subtlety may explain the many contradictory studies present within the DNA methylation and nucleosome stability field.

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